nature portfolio

Peer Review File

Spatial patterning of fibroblast TGFβ signaling underlies treatment resistance in rheumatoid arthritis

Corresponding Author: Dr Kevin Wei

This file contains all editorial decision letters in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
The study by Bhamidipati et al. identifies a population of fibroblasts, defined by high expression of COMP that is associated with treatment resistant, non-remission rheumatoid arthritis. These cells were identified by spatial transcriptomic analysis to be highly fibrogenic, by expression of extracellular matrix transcripts, and localized to perivascular spaces. Within these perivascular niches, two populations of fibroblasts were identified, COMPhi and POSTNhi fibroblasts, which differed in their proximity to endothelial cells. POSTNhi were in closer proximity to ECs, while COMPhi fibroblasts were in the distal regions of the perivascular spaces. This study aims to link proximity to ECs, through the production of gradients TGFb signalling to this observed fibroblast patterning in the perivascular spaces of non-remission RA. This link was explored through a combination of spatial transcriptomic analyses and in vitro approaches that link Notch signalling in endothelial cells to the maintenance of the POSTNhi and COMPhi fibroblast phenotypes through mediation of TGFb and TGFb receptors. Overall, this study provides an interesting look into the fibroblast landscape of non-remission RA which could inform new therapeutic avenues for a treatment resistant disease. Yet, in its current form the submitted work is largely descriptive. Inclusion of in vivo models or clinical human datasets (as they foreshadow in the discussion) to situate the potentially novel findings in the context of ameliorating RA disease progression is needed to elicit significant enthusiasm.
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• The authors do a great job of identifying the different niches found in the synovium, however the link between the niches and non-remitting patients is difficult to visualize in figure 1. Data that solidifies the connection between increased fibrogenic niches and non-remission RA is crucial to this story and is lacking in Figure 1.
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observed since while they are both downstream from Notch-TGFβ, they are independently regulated (not anticorrelated in cartilage expression. Are the authors proposing a new mechanism?

In the discussions on notch and TGFβ in co-culture experiments, there is significant discussion on the isoforms of TGFβ, while there is no discussion on notch isoforms. Based on prior literature, Notch3 is the isoform that is upstream of COMP and POSTN expression. Did the authors explore which Notch isoform is expressed and which isoform controls the TGFβ pathway? Also, the discussion on the TGFb1, 2, 3 is confusing. It would be good to summarize this in a pathway schematic. Unless this reviewer is mistaken, DLL4, which induces TGFβ1 is a ligand for Notch 1 and Notch 2 but not Notch 3. Does this mean Notch 3 is not involved in the fibrogenic process in non-remission? This is confusing. Also, the Notch signaling pathway can regulate COMP protein production, according to studies on cartilage development. Specifically, the Notch pathway influences the differentiation of mesenchymal progenitor cells (MPCs), which participate in cartilage formation, and thus can impact COMP protein expression. However, are the fibrogenic fibroblasts these mesenchymal cells? Some clear discussion on this is merited.

Does the discussion on the distance of COMPhi cell being higher in DLL4 knockdown refer to EC proximity? Figs 6e and 6f do not seem to clearly state this point.

In the discussion of COMPhi regions, it is stated they become pauci-cellular (Fig 7e, g). Does this refer to pauci-immune or pauci-fibroblast? Again, connecting this observation to niches in post-treatment responsive and nonresponsive patients would help.

Reviewer #3

(Remarks to the Author)
Summary:
In this study, Bhamidipati, McIntyre, et al use spatial transcriptomic and proteomic profiling of synovial biopsies from patients with rheumatoid arthritis to explore potential pathogenic mechanisms of inadequate clinical response to standard of care anti-inflammatory treatment. They identify spatially distinct subpopulations of synovial fibroblasts and associate the expansion of perivascular fibroblasts expressing high levels of COMP, a matricellular protein, with treatment resistance irrespective of effects on inflammatory cell infiltrates. They then use a combination of transcriptomic analyses from other synovial biopsy datasets and several in vitro models to explore a gradient of TGFb signaling regulated by endothelial cell-fibroblast Notch signaling and conclude that isoform-selective TGFb targeting may offer potential benefit to RA patients independent of anti-inflammatory treatment. This study is intriguing in that it endeavors to provide mechanistic insights beyond the largely descriptive work that has preceded it, and uses an impressive complement of analytical technologies, primary patient-derived samples, pre-existing datasets, and model systems to explore its hypotheses. However, there are several instances where clarification and/or additional experimentation would help in interpreting data and substantiating conclusions.

Specific comments:
1) Clinical: the spatial transcriptomic data are derived from a small cohort of 17 patients in the 396.10 study classified according to DAS28-ESR remission status (supplementary table 1). A more complete description of the clinical outcomes would aid in interpretation of the biopsy data. Given that the DAS score is a composite measure of swollen and tender joint counts, inflammatory biomarkers, and a subjective report of pain/disability, it would be helpful to understand which of those components drove remission or non-remission in this study. Of note, at baseline, the “non-remission” patients appear to have higher DAS scores than the “remission” patients, but is this because they have more active joints, more inflammatory biomarkers, more pain, or some combination thereof? Remission is defined as a 6-month DAS28-ESR score <2.6 but it’s not clear whether the “non-remission” patients experienced any decrease in DAS score on treatment (but remained >2.6) or had different patterns of response from the “remission” patients. For example, in the PEAC cohort (figure 1f), the definition of “non-response” is fairly strict: DAS28-CRP > 5.1 and improvement < 0.6. In 396.10, did “non-remission” patients with a decrease in inflammatory infiltrates in their synovial biopsies exhibit meaningful changes in joint counts or ESR/CRP but not pain? In the Discussion section (lines 523-527), the authors lean heavily on “non-inflammatory pain” as being a driver of incomplete response to anti-inflammatory treatment; is this concept applicable specifically to the patients examined in the 396.10 study? What evidence is present in this study that “non-inflammatory pain” is indeed linked to fibrogenic fibroblast populations?

2) More detail on biopsies is needed. How do we know the associations between spatial transcriptomic assessments and clinical outcomes over time are based on “representative” biopsies of “representative” joints and not simply due to sampling artifacts? Which joints were biopsied, were they selected via ultrasound assessment for active synovitis, and were the same or different joints assessed at baseline and 6 months? Were any blood biomarkers related to patterns of synovial pathology (e.g., as described for patients in the AMP cohort in figure 7i) measured in the patients from the 396.10 study from which these biopsies and clinical characterization were derived? It would be interesting to see whether elevated serum COMP (and/or other fibroblast-derived biomarkers) at baseline and/or pharmacodynamic effects on those markers had any association with clinical response in the 396.10 patients.

3) This study raises many questions about TGFbeta signaling: via several in vitro mechanistic models, the authors conclude

that the main effect of endothelial-derived Notch signals on fibroblasts is the spatial regulation of TGFBR3 expression (lines 435-440). While there are differential effects on TGFb isoform expression, it appears that TGFb1 and b3 are more significantly affected than TGFb2, and the authors suggest that TGFb3 may be the isoform responsible for the COMPhi fibroblast population expansion. However, relatively scant attention is paid to the previously described roles for TGFb receptor subunits and selectivity for isoforms: whereas TGFb1 and b3 bind strongly to TGFBR2 and recruit TGFBR1 (ALK5) to induce intracellular signaling, TGFb2 has lower affinity for TGFBR1 and 2 (e.g., PMID 20207738). Many studies have suggested that TGFBR3 specifically acts to facilitate TGFb2's ability to bind to TGFBR1/2 complexes and direct signaling, whereas its effects on other TGFb isoforms are minimal. Is it therefore possible that TGFb2, rather than TGFb3, is the potential therapeutic target? In figure 4j, siRNA knockdown of each TGFb isoform individually (including TGFb2) in DLL4-treated cells shows comparable modest reductions in pro-Col1A1 production despite no differential effect of DLL4 +/- DAPT treatment on TGFb2 expression (figure 4e), but less reduction than knocking down either TGFbR2 or R3. Why is there no effect when TGFbR1 is knocked down (given that ALK5 is required for SMAD activation and signaling, e.g., in figure 4i, where an ALK5 inhibitor is used)? While there are data to show the efficiency of DLL4 knockdown (extended data figure 5), there is not similar confirmation of the knockdown efficiency of the other targets, which may be a simple explanation for the divergent results. Given the tight regulation of TGFb positioning by milieu molecules (e.g., LTBRs, LRRCs) and activation of latent TGFb (e.g., by integrin binding or LAP cleavage), do the authors believe that TGFb signaling is autocrine, paracrine (i.e., dependent on close cell-cell contact), or via broader diffusion of soluble active TGFb? This in turn may be highly dependent on which isoform is being considered: whereas latent TGFb1 has a very high threshold for activation, latent TGFb2 and TGFb3 have intrinsic activity.

4) There are several different in vitro models employed in figures 3-6 but it is confusing why certain models were used for certain purposes. It appears that the organoid model data in figure 4a-d are derived from a dataset in a previous publication from this group (ref. 9) but the "new" experiments in this study are in a simpler 2-D co-culture system with HUVECs and synovial fibroblasts. I have no issues with recycling data from a prior study but am curious whether the patterns of gene expression and cellular positioning identified in the 3-D organoids and the 2-D culture system align with each other. In particular, the role of milieu molecules and ECM in positioning and activating TGFb isoforms (see comment #3, above) may be different across these experimental systems and/or differentially relevant to an in vivo setting. Additionally, the pre/post treatment synovial biopsy data (figure 7c-d) suggest that DMARD treatment, but not anti-TNF treatment, is associated with an increased percentage of COMPhi fibroblasts in vivo. Do any or all of the DMARDs have direct effects on readouts in the in vitro EC-FB coculture systems (i.e., absent an immune cell component)? Does the presence or absence of TNF in vitro affect Notch or TGFb signaling and/or fibroblast phenotypes?

5) Biomarkers: as noted in comment #2 (above), were serum samples available for blood biomarker analysis in the 396.10 study? If so, are the effects consistent with the data from the AMP study in figure 7i? Olink measures over 5000 protein analytes but it's not clear whether the indicated biomarkers in 7i are the most significantly different between CTAP groups among all the analytes or were cherry-picked to make a particular point (perhaps a volcano plot of all the analytes would be instructive here or in a supplementary figure). Why is periostin not shown? Considering that all the characterization of fibroblast subpopulations in the spatial data and the well-established use of blood periostin as a biomarker in multiple inflammatory and fibrotic conditions (it is also on the Olink panel and present in the linked raw data file), this seems like a curious omission. Given the mechanistic and spatial data elsewhere in this study (discussed in comment #3, above), one could speculate that periostin in POSTNhi fibroblasts is more TGFb3 (+/- TGFb1) dependent whereas COMP in COMPhi fibroblasts is more TGFb2/TGFBR3 dependent. Ultimately, if the findings in this study are to have clinical utility, noninvasive biomarkers will likely be needed to enable and interpret future interventional studies.

6) Recommend biostatistical review: in the methods section, it is stated that p-values are not adjusted for multiple testing; given the number of hypotheses tested in several of the experiments, is this appropriate? For example, in the analysis of the Olink biomarker data (see comment #5, above), the "unadjusted" p-value for serum COMP in CTAP-EFM vs. other is about 0.01 according to the color map. Even assuming that the analytes shown were rationally pre-specified and limited, there are 10 of them, thus a simple Bonferroni adjustment would yield a "non-significant" p-value according to the Methods section ("a p-value < 0.05 was considered statistically significant"). In the analyses of spatial data, there are some staggeringly low p-values (e.g., 10e-308 in figure 6n), which may be plausible given that >10e4 cells were imaged; nevertheless, the substantial overlap in distributions calls into question whether these highly "significant" p-values actually represent biologically meaningful differences and whether the statistical methods employed are appropriate for this type of data.

Decision Letter:

1st May 2025

Dear Kevin,

Thank you for providing a point-by-point response to the referees' comments on your manuscript entitled, "Spatial patterning of fibroblast TGFβ signaling underlies treatment resistance in rheumatoid arthritis". As noted previously, while they find your work of considerable potential interest, they have raised quite substantial concerns that must be addressed. In light of these comments, we cannot accept the current manuscript for publication, but would be very interested in considering a revised version that addresses these serious concerns along the lines proposed in your point-by-point rebuttal.

We invite you to submit a substantially revised manuscript, however please bear in mind that we will be reluctant to approach the referees again in the absence of major revisions.

Specifically, the revision should include new experiments to address:

(1) generate RA patient-derived synovial organoid cultures from treatment-resistant patients to test the effects of Notch and TFGbeta inhibitors on gene/protein expression, and and test their ability to ablate fibrosis in synovial tissue

(2) to determine which TGFbeta signaling components are relevant in the RA disease process, perform systematic siRNA-mediated knockdown (and/or CRISPR/Cas9-mediated deletion) of TGFbeta isoforms and receptors to examine effects on synovial fibroblast expression of extracellular matrix proteins in response to DLL4-Notch signaling

(3) perform in vitro experiments to determine whether TGF-beta signaling in synovial fibroblasts is dependent upon cell-cell contact or sensitive to soluble released TGF-beta

(4) compare the 2D and 3D-organoid in vitro culture systems to determine if the spatial patterning of TGF-beta activity is consistent in the two culture systems; comparison of the same 50-gene custom transcript panel, and responses to Notch and TGF-beta inhibitors (protein expression and fibrogenic activation)

(5) examine the effects of disease-modifying anti-rheumatic drugs and TNF (inhibitors) on DLL4- or TGFb3-stimulated fibroblasts, assaying for inflammatory cytokines and fibrogenic activation

(6) perform OLINK proteomic analysis on biobanked patient serum to test the association between fibroblast-derived proteins and fibrogenic niches and patient treatment outcomes

Please include the additional textual clarifications, validation experiments and reanalysis of existing datasets as indicated in your response letter.

When you revise your manuscript, please take into account all reviewer and editor comments, please highlight all changes in the manuscript text file in Microsoft Word format.

We are committed to providing a fair and constructive peer-review process. Do not hesitate to contact us if there are specific requests from the reviewers that you believe are technically impossible or unlikely to yield a meaningful outcome.

When revising your manuscript:

* Include a “Response to referees” document detailing, point-by-point, how you addressed each referee comment. If no action was taken to address a point, you must provide a compelling argument. This response will be sent back to the referees along with the revised manuscript.

* If you have not done so already please begin to revise your manuscript so that it conforms to our Article format instructions at http://www.nature.com/ni/authors/index.html. Refer also to any guidelines provided in this letter.

* Include a revised version of any required reporting checklist. It will be available to referees (and, potentially, statisticians) to aid in their evaluation if the manuscript goes back for peer review. A revised checklist is essential for re-review of the paper.

The Reporting Summary can be found here:
https://www.nature.com/documents/nr-reporting-summary.pdf

When submitting the revised version of your manuscript, please pay close attention to our href="https://www.nature.com/nature-portfolio/editorial-policies/image-integrity">Digital Image Integrity Guidelines. and to the following points below:

-- that unprocessed scans are clearly labelled and match the gels and western blots presented in figures.
-- that control panels for gels and western blots are appropriately described as loading on sample processing controls
-- all images in the paper are checked for duplication of panels and for splicing of gel lanes.

Extended Data figures and tables are online-only (appearing in the online PDF and full-text HTML version of the paper), peer-reviewed display items that provide essential background to the Article but are not included in the printed version of the paper due to space constraints or being of interest only to a few specialists. A maximum of ten Extended Data display items (figures and tables) is typically permitted. When re-submitting your manuscript, please ensure that any supplementary figures and tables that are more critical to the manuscript’s conclusions are converted to Extended data to increase these data’s visibility.

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Please do not hesitate to contact me if you have any questions or would like to discuss the required revisions further.

Thank you for the opportunity to review your work.

Sincerely,

Laurie A. Dempsey, Ph.D.
Senior Editor
Nature Immunology
l.dempsey@us.nature.com
ORCID: 0000-0002-3304-796X

Referee expertise:

Referee #1: immune-stromal cell interactions

Referee #2: multi-OMIC systems analysis

Referee #3: human autoimmune/autoinflammatory disease

Reviewers' Comments:

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In the discussions on notch and TGFβ in co-culture experiments, there is significant discussion on the isoforms of TGFβ, while there is no discussion on notch isoforms. Based on prior literature, Notch3 is the isoform that is upstream of COMP and POSTN expression. Did the authors explore which Notch isoform is expressed and which isoform controls the TGFβ pathway? Also, the discussion on the TGFb1, 2, 3 is confusing. It would be good to summarize this in a pathway schematic. Unless this reviewer is mistaken, DLL4, which induces TGFβ1 is a ligand for Notch 1 and Notch 2 but not Notch 3. Does this mean Notch 3 is not involved in the fibrogenic process in non-remission? This is confusing. Also, the Notch signaling pathway can regulate COMP protein production, according to studies on cartilage development. Specifically, the Notch pathway influences the differentiation of mesenchymal progenitor cells (MPCs), which participate in cartilage formation, and thus can impact COMP protein expression. However, are the fibrogenic fibroblasts these mesenchymal cells? Some clear discussion on this is merited.

Does the discussion on the distance of COMPhi cell being higher in DLL4 knockdown refer to EC proximity? Figs 6e and 6f do not seem to clearly state this point.

In the discussion of COMPhi regions, it is stated they become pauci-cellular (Fig 7e, g). Does this refer to pauci-immune or pauci-fibroblast? Again, connecting this observation to niches in post-treatment responsive and nonresponsive patients would help.

Reviewer #3 (Remarks to the Author):

Summary:
In this study, Bhamidipati, McIntyre, et al use spatial transcriptomic and proteomic profiling of synovial biopsies from patients with rheumatoid arthritis to explore potential pathogenic mechanisms of inadequate clinical response to standard of care anti-inflammatory treatment. They identify spatially distinct subpopulations of synovial fibroblasts and associate the expansion of perivascular fibroblasts expressing high levels of COMP, a matricellular protein, with treatment resistance irrespective of effects on inflammatory cell infiltrates. They then use a combination of transcriptomic analyses from other synovial biopsy datasets and several in vitro models to explore a gradient of TGFb signaling regulated by endothelial cell-fibroblast Notch signaling and conclude that isoform-selective TGFb targeting may offer potential benefit to RA patients independent of anti-inflammatory treatment. This study is intriguing in that it endeavors to provide mechanistic insights beyond the largely descriptive work that has preceded it, and uses an impressive complement of analytical technologies, primary patient-derived samples, pre-existing datasets, and model systems to explore its hypotheses. However, there are several instances where clarification and/or additional experimentation would help in interpreting data and substantiating conclusions.

Specific comments:
1) Clinical: the spatial transcriptomic data are derived from a small cohort of 17 patients in the 396.10 study classified according to DAS28-ESR remission status (supplementary table 1). A more complete description of the clinical outcomes would aid in interpretation of the biopsy data. Given that the DAS score is a composite measure of swollen and tender joint counts, inflammatory biomarkers, and a subjective report of pain/disability, it would be helpful to understand which of those components drove remission or non-remission in this study. Of note, at baseline, the “non-remission” patients appear to have higher DAS scores than the “remission” patients, but is this because they have more active joints, more inflammatory biomarkers, more pain, or some combination thereof? Remission is defined as a 6-month DAS28-ESR score <2.6 but it’s not clear whether the “non-remission” patients experienced any decrease in DAS score on treatment (but remained >2.6) or had different patterns of response from the “remission” patients. For example, in the PEAC cohort (figure 1f), the definition of “non-response” is fairly strict: DAS28-CRP > 5.1 and improvement < 0.6. In 396.10, did “non-remission” patients with a decrease in inflammatory infiltrates in their synovial biopsies exhibit meaningful changes in joint counts or ESR/CRP but not pain? In the Discussion section (lines 523-527), the authors lean heavily on “non-inflammatory pain” as being a driver of

incomplete response to anti-inflammatory treatment; is this concept applicable specifically to the patients examined in the 396.10 study? What evidence is present in this study that “non-inflammatory pain” is indeed linked to fibrogenic fibroblast populations?

2) More detail on biopsies is needed. How do we know the associations between spatial transcriptomic assessments and clinical outcomes over time are based on “representative” biopsies of “representative” joints and not simply due to sampling artifacts? Which joints were biopsied, were they selected via ultrasound assessment for active synovitis, and were the same or different joints assessed at baseline and 6 months? Were any blood biomarkers related to patterns of synovial pathology (e.g., as described for patients in the AMP cohort in figure 7i) measured in the patients from the 396.10 study from which these biopsies and clinical characterization were derived? It would be interesting to see whether elevated serum COMP (and/or other fibroblast-derived biomarkers) at baseline and/or pharmacodynamic effects on those markers had any association with clinical response in the 396.10 patients.

3) This study raises many questions about TGFbeta signaling: via several in vitro mechanistic models, the authors conclude that the main effect of endothelial-derived Notch signals on fibroblasts is the spatial regulation of TGFBR3 expression (lines 435-440). While there are differential effects on TGFb isoform expression, it appears that TGFb1 and b3 are more significantly affected than TGFb2, and the authors suggest that TGFb3 may be the isoform responsible for the COMPhi fibroblast population expansion. However, relatively scant attention is paid to the previously described roles for TGFb receptor subunits and selectivity for isoforms: whereas TGFb1 and b3 bind strongly to TGFBR2 and recruit TGFBR1 (ALK5) to induce intracellular signaling, TGFb2 has lower affinity for TGFBR1 and 2 (e.g., PMID 20207738). Many studies have suggested that TGFBR3 specifically acts to facilitate TGFb2’s ability to bind to TGFBR1/2 complexes and direct signaling, whereas its effects on other TGFb isoforms are minimal. Is it therefore possible that TGFb2, rather than TGFb3, is the potential therapeutic target? In figure 4j, siRNA knockdown of each TGFb isoform individually (including TGFb2) in DLL4-treated cells shows comparable modest reductions in pro-Col1A1 production despite no differential effect of DLL4 +/- DAPT treatment on TGFb2 expression (figure 4e), but less reduction than knocking down either TGFbR2 or R3. Why is there no effect when TGFbR1 is knocked down (given that ALK5 is required for SMAD activation and signaling, e.g., in figure 4i, where an ALK5 inhibitor is used)? While there are data to show the efficiency of DLL4 knockdown (extended data figure 5), there is not similar confirmation of the knockdown efficiency of the other targets, which may be a simple explanation for the divergent results. Given the tight regulation of TGFb positioning by milieu molecules (e.g., LTBRs, LRRCs) and activation of latent TGFb (e.g., by integrin binding or LAP cleavage), do the authors believe that TGFb signaling is autocrine, paracrine (i.e., dependent on close cell-cell contact), or via broader diffusion of soluble active TGFb? This in turn may be highly dependent on which isoform is being considered: whereas latent TGFb1 has a very high threshold for activation, latent TGFb2 and TGFb3 have intrinsic activity.

4) There are several different in vitro models employed in figures 3-6 but it is confusing why certain models were used for certain purposes. It appears that the organoid model data in figure 4a-d are derived from a dataset in a previous publication from this group (ref. 9) but the “new” experiments in this study are in a simpler 2-D co-culture system with HUVECs and synovial fibroblasts. I have no issues with recycling data from a prior study but am curious whether the patterns of gene expression and cellular positioning identified in the 3-D organoids and the 2-D culture system align with each other. In particular, the role of milieu molecules and ECM in positioning and activating TGFb isoforms (see comment #3, above) may be different across these experimental systems and/or differentially relevant to an in vivo setting. Additionally, the pre/post treatment synovial biopsy data (figure 7c-d) suggest that DMARD treatment, but not anti-TNF treatment, is associated with an increased percentage of COMPhi fibroblasts in vivo. Do any or all of the DMARDs have direct effects on readouts in the in vitro EC-FB coculture systems (i.e., absent an immune cell component)? Does the presence or absence of TNF in vitro affect Notch or TGFb signaling and/or fibroblast phenotypes?

5) Biomarkers: as noted in comment #2 (above), were serum samples available for blood biomarker analysis in the 396.10 study? If so, are the effects consistent with the data from the AMP study in figure 7i? Olink measures over 5000 protein analytes but it’s not clear whether the indicated biomarkers in 7i are the most significantly different between CTAP groups among all the analytes or were cherry-picked to make a particular point (perhaps a volcano plot of all the analytes would be instructive here or in a supplementary figure). Why is periostin not shown? Considering that all the characterization of fibroblast subpopulations in the spatial data and the well-established use of blood periostin as a biomarker in multiple inflammatory and fibrotic conditions (it is also on the Olink panel and present in the linked raw data file), this seems like a curious omission. Given the mechanistic and spatial data elsewhere in this study (discussed in comment #3, above), one could speculate that periostin in POSTNhi fibroblasts is more TGFb3 (+/- TGFb1) dependent whereas COMP in COMPhi fibroblasts is more TGFb2/TGFBR3 dependent. Ultimately, if the findings in this study are to have clinical utility, noninvasive biomarkers will likely be needed to enable and interpret future interventional studies.

6) Recommend biostatistical review: in the methods section, it is stated that p-values are not adjusted for multiple testing; given the number of hypotheses tested in several of the experiments, is this appropriate? For example, in the analysis of the Olink biomarker data (see comment #5, above), the “unadjusted” p-value for serum COMP in CTAP-EFM vs. other is about 0.01 according to the color map. Even assuming that the analytes shown were rationally pre-specified and limited, there are 10 of them, thus a simple Bonferroni adjustment would yield a “non-significant” p-value according to the Methods section (“a p-value < 0.05 was considered statistically significant”). In the analyses of spatial data, there are some staggeringly low p-values (e.g., 10e-308 in figure 6n), which may be plausible given that >10e4 cells were imaged; nevertheless, the substantial overlap in distributions calls into question whether these highly “significant” p-values actually represent biologically meaningful differences and whether the statistical methods employed are appropriate for this type of data.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
The authors have adequately addressed my concerns.

Reviewer #2

(Remarks to the Author)
Review of revised version,
NI-A39921 "Spatial patterning of fibroblast TGFβ signaling underlies treatment resistance in rheumatoid arthritis"

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spatial transcriptomics data between pre- and post-treatment and those who are non-remission in contrast to remission, warrants more analysis to justify the authors' conclusions. This reviewer refers to the critiques above for the authors to consider.

Reviewer #3

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In this revised manuscript, Bhamidipati, McIntyre, et al have made extensive revisions, included additional information, and added substantial experimental data to strengthen their case for a dysregulated synovial fibroblast population characterized by Notch- and TGFb-dependent signaling gradients in treatment-refractory rheumatoid arthritis.

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7) Thank you for the additional context on statistical methods.

Decision Letter:

6th Oct 2025

Dear Dr Wei,

Thank you for providing a point-by-point response to the remaining concerns voiced by referees #2 and #3 on your manuscript entitled, "Spatial patterning of fibroblast TGFβ signaling underlies treatment resistance in rheumatoid arthritis". You will see from their comments below that while they find your work of interest, some important points are raised. We are very interested in the possibility of publishing your study in Nature Immunology, along the lines suggested in your recent response (6th Oct 2025) to these concerns in the form of a revised manuscript before we make a final decision on publication.

It seems as if much of the revision can be performed by further analysis of data in-hand. We therefore invite you to revise your manuscript taking into account all reviewer comments. Please highlight all changes in the manuscript text file in Microsoft Word format.

We are committed to providing a fair and constructive peer-review process. Do not hesitate to contact us if there are specific requests from the reviewers that you believe are technically impossible or unlikely to yield a meaningful outcome.

When revising your manuscript:

* Include a “Response to referees” document detailing, point-by-point, how you addressed each referee comment. If no action was taken to address a point, you must provide a compelling argument. This response will be sent back to the referees along with the revised manuscript.

* If you have not done so already please begin to revise your manuscript so that it conforms to our Article format instructions at http://www.nature.com/ni/authors/index.html. Refer also to any guidelines provided in this letter.

* Please include a revised version of any required reporting checklist. It will be available to referees to aid in their evaluation of the manuscript goes back for peer review. They are available here:

Reporting summary:
https://www.nature.com/documents/nr-reporting-summary.pdf

When submitting the revised version of your manuscript, please pay close attention to our href="https://www.nature.com/nature-portfolio/editorial-policies/image-integrity">Digital Image Integrity Guidelines. and to the following points below:

-- that unprocessed scans are clearly labelled and match the gels and western blots presented in figures.
-- that control panels for gels and western blots are appropriately described as loading on sample processing controls
-- all images in the paper are checked for duplication of panels and for splicing of gel lanes.

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Finally, please ensure that you retain unprocessed data and metadata files after publication, ideally archiving data in perpetuity, as these may be requested during the peer review and production process or after publication if any issues arise.

Please use the link below to submit your revised manuscript and related files:

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We hope to receive your revised manuscript within four weeks. If you cannot send it within this time, please let us know. We will be happy to consider your revision so long as nothing similar has been accepted for publication at Nature Immunology or published elsewhere.

Please do not hesitate to contact me if you have any questions or would like to discuss these revisions further.

Nature Immunology is committed to improving transparency in authorship. As part of our efforts in this direction, we are now requesting that all authors identified as ‘corresponding author’ on published papers create and link their Open Researcher and Contributor Identifier (ORCID) with their account on the Manuscript Tracking System (MTS), prior to acceptance. ORCID helps the scientific community achieve unambiguous attribution of all scholarly contributions. You can create and link your ORCID from the home page of the MTS by clicking on ‘Modify my Springer Nature account’. For more information please visit please visit www.springernature.com/orcid.

We look forward to seeing the revised manuscript and thank you for the opportunity to review your work.

Kind regards,

Laurie

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ORCID: 0000-0002-3304-796X

Reviewers' Comments:

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The authors have adequately addressed my concerns.

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Decision Letter:

Our ref: NI-A39921B

29th Oct 2025

Dear Kevin,

Thank you for submitting your revised manuscript "Spatial patterning of fibroblast TGFβ signaling underlies treatment resistance in rheumatoid arthritis" (NI-A39921B). It has now been seen by the original referees #2 and #3 and their comments are below. The reviewers find that the paper has improved in revision, and therefore we'll be happy in principle to publish it in Nature Immunology, pending minor revisions to satisfy the referees' final requests and to comply with our editorial and formatting guidelines.

We will now perform detailed checks on your paper and will send you a checklist detailing our editorial and formatting requirements in about a week. Please do not upload the final materials and make any revisions until you receive this additional information from us.

If you had not uploaded a Word file for the current version of the manuscript, we will need one before beginning the editing process; please email that to immunology@us.nature.com at your earliest convenience.

Thank you again for your interest in Nature Immunology Please do not hesitate to contact me if you have any questions.

Kind regards,

Laurie

Laurie A. Dempsey, Ph.D.
Senior Editor
Nature Immunology
l.dempsey@us.nature.com
ORCID: 0000-0002-3304-796X

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Reviewer #3 (Remarks to the Author):

The latest revision of the manuscript adequately addresses my concerns. Thanks to the authors for their diligence in revising and updating the work.

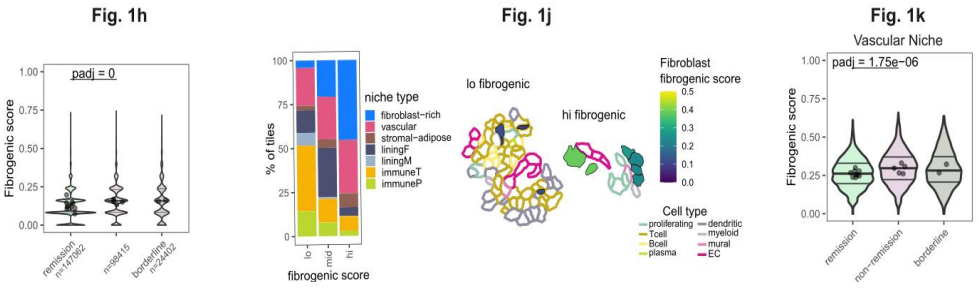
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We thank the reviewers for their time and effort in considering our manuscript, **NI-A39921** "**Spatial patterning of fibroblast TGFβ signaling underlies treatment resistance in rheumatoid arthritis**" for publication in Nature Immunology. We are delighted to see positive feedback from the reviewers. Reviewer 1 (R1) commented that our "*study provides an interesting look into the fibroblast landscape of non-remission RA, which could inform new therapeutic avenues for a treatment-resistant disease*" and reviewer 2 (R2) described the study as having "*several interesting aspects*". Further, reviewer 3 (R3) found our study *"intriguing in that it endeavors to provide mechanistic insights beyond the largely descriptive work that has preceded it, and uses an impressive complement of analytical technologies, primary patient-derived samples, pre-existing datasets, and model systems to explore its hypotheses*." We also appreciate the constructive feedback from the three reviewers and the opportunity to improve our study. In our revised manuscript, we provide new experimental data and in-depth analyses to address each point raised by the reviewers. We also substantially improved the organization and presentation of the data to enhance the narrative. Please see our point-by-point response to the reviewers' comments below.

Reviewer #1

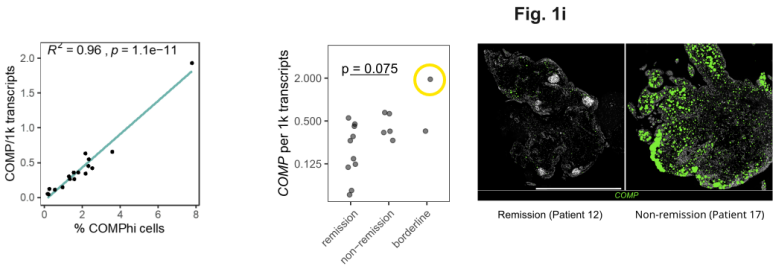
R1C1. ***"****The authors do a great job of identifying the different niches found in the synovium, however the link between the niches and non-remitting patients is difficult to visualize in figure 1. Data that solidifies the connection between increased fibrogenic niches and non-remission RA is crucial to this story and is lacking in Figure 1."*

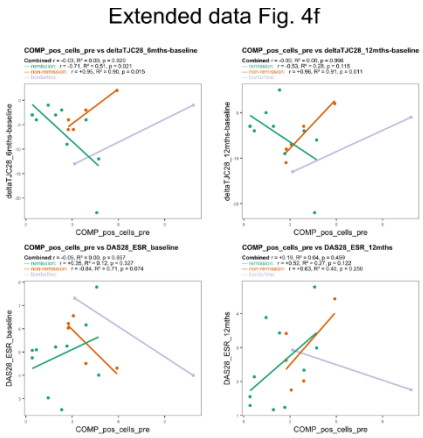
We thank the reviewer for their comment and agree that emphasizing the connection between fibrogenic niches and non-remission in Figure 1 is essential. We have amended **Fig. 1** to highlight the increased fibrogenic signaling in the baseline biopsies of non-remission patients and have added plots to visualize fibrogenic niches. First, we added a violin plot representing the distribution of single-cell and patient-level fibroblast fibrogenic signature scores across the 3 remission groups (**Fig. 1h)**; we show that the fibrogenic activation score is significantly elevated among fibroblasts from non-remitting patients. To identify the specific niche in which the fibrogenic signature was most elevated, we classified all niche tiles as high, medium and low for the fibrogenic signature and provide visual examples highlighting the composition of low-fibrogenic (immune-dominant) and high fibrogenic (stromal-dominant) niches (**Fig. 1j**). Further, we identified vascular niches as the site where the fibrogenic signature was strongly enriched in non-remission RA (**Fig. 1k)**. The updated analysis at the end of **Fig. 1** supports our hypothesized link between refractory RA and the perivascular origin of synovial tissue fibrosis. Further, this allows a clearer progression to describing perivascular compartmentalization of TGFβ signaling as presented in **Fig. 2**.



R1C2. "*Readers would benefit by a clearer focus on COMPhi cells (vs. POSTN+ cells) as biomarkers for non-remitting RA (Figs. 2 and 7 in particular). Clarity on the following points would be appreciated: Are COMPhi cells absent in remission RA? Can these cells be quantified? Transcript number is informative, but cell quantification would be of great benefit. Are there fibroblast phenotypes in the literature that overlap with COMPhi cells? Is the number of COMP transcripts associated with higher disease severity or only non-response to therapeutics?*

We thank the reviewer for recommending further clarity regarding the use of *COMP* expression as a biomarker of non-remission. To address this, we have performed deeper clinical analyses of the 396.10 cohort in relation to *COMP*+ cell abundance and have substantially re-organized **Fig. 1** and associated **Extended Data Fig. 4**. Firstly, we evaluated the correlation between normalized *COMP* transcript levels and *COMP*hi cell abundance (defined as a cell for which *COMP* is a top 50 marker gene) and found a very strong correlation (r = 0.98, p = 1.1e-11), suggesting that the metrics are concordant with each other. For the manuscript, we have chosen to show normalized transcript abundance (**Fig. 1i**) because it is a more sensitive measure of sample-level differences, as it is not prone to transcript loss that may results from cell segmentation errors. We would like to note that our transcript quantification plot (**Fig. 1i**) has changed due to an error in our initial patient classifications; in looking more closely at the clinical metadata, we realized that we erroneously classified a borderline patient (6 month DAS28-ESR 2.26 < 2.6 cutoff, 6 month DAS28-CRP 2.58 > 2.4 cutoff) as a non-remission patient which we have now corrected throughout the manuscript with the inclusion of a "borderline" category in plots that stratify patients by remission status. As can be observed from the plot, *COMP* is also expressed in remission patients though at lower average levels. To further detail the association between *COMP*+ cell abundance and measures of disease severity, we have added 4 correlation plots highlighting key clinical observations. (**Extended data Fig. 4g**). Among non-remission patients, we found that *COMP*+ cell abundance correlated with lesser improvements in tender joint count (deltaTJC28 at 6 months and 12 months.) We also found that, *COMP*+ cell abundance shows a positive association with DAS28-ESR at 12 months, but not at baseline, reflecting the link between baseline *COMP*+ cell abundance and long-term disease activity. Lastly, *COMP* has been explicitly described as a marker for expanded fibrogenic fibroblast subpopulations in scleroderma, which we have updated the references to reflect (References **33,34**).



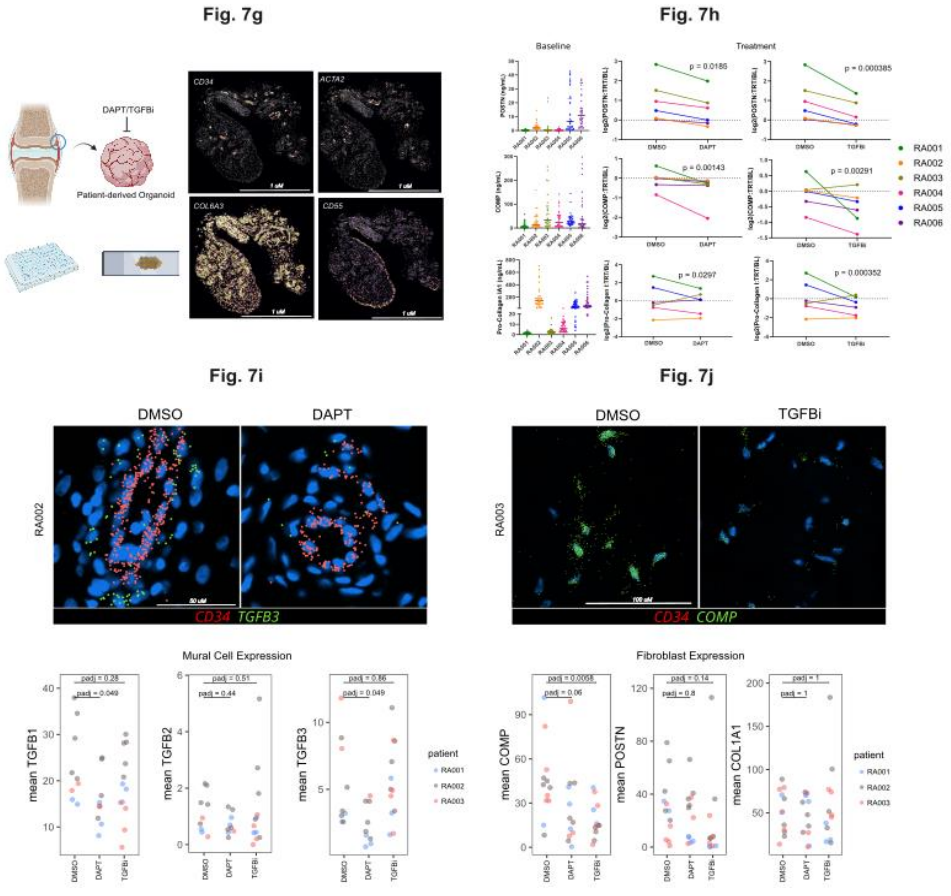


R1C3. "*While the link between Notch signalling and TGFb isoforms and receptors is displayed well through your in vitro approaches, the causal link between endothelial cells and your perivascular fibroblast phenotype/function in disease models is lacking.*"

We strongly agree with the reviewer's comment regarding the importance of linking *in vitro* defined mechanisms to disease-relevant model systems. While several mouse models of arthritis exist, each model differs in their ability to model different aspects of RA pathology, such as

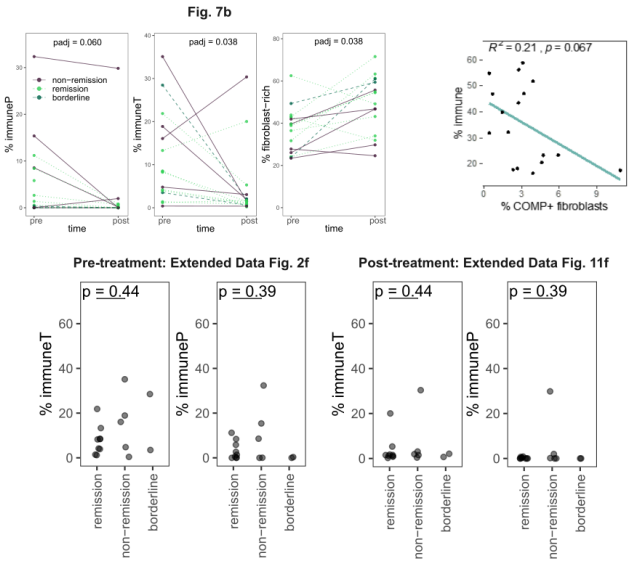
autoimmunity, innate immune response, synovial fibroblast activation, and bone erosion. To our knowledge, none of these models faithfully replicates synovial tissue pathology in treatment-resistant RA. In particular, the presence of fibrogenic niches and their role in mouse arthritis pathology has not been established; therefore, it is unclear the extent to which interventions in mouse arthritis models would provide substantive functional evidence regarding RA treatment resistance in human disease. The reviewer highlights a major unmet need in that our field needs an animal model that faithfully recapitulates the diversity of synovial tissue pathology in RA patients, as our consortium AMP RA/SLE recently described (Zhang et al., Nature 2023). This is an enormous challenge for our field and outside the scope of our current study.

To overcome this challenge and to address the functional relevance of TGFβ and Notch signaling on synovial tissue fibrogenesis in RA, we have developed a novel RA patient-derived organoid (PDO) approach, using synovial tissue from RA patients (n = 6) who underwent synovectomy or arthroplasty. Importantly, our RA PDOs maintain patients' original synovial tissue microarchitecture and cellular composition, thus providing a system by which we can examine the effect of disruption Notch and TGFB signaling in RA synovial tissue *ex vivo*. To do this, we derived >30 PDOs from each donor RA synovial tissue, using a 2um punch biopsy, and embedded them in Matrigel overnight in low-serum media that has been optimized to maintain synovial endothelial cells and fibroblasts. We then treated PDOs with inhibitors against Notch or TGFβ signaling, followed by measuring supernatant fibrogenic protein production and then profiling the organoids with spatial transcriptomics to determine changes in patterning of gene expression (**Fig. 7g-j, Extended Data Fig. 13**). Treatment of PDOs with Notch or TGFβ inhibitors led to marked reductions in supernatant Pro-collagent 1, COMP and POSTN protein levels (**Fig. 7h**). Further, we applied spatial transcriptomics to profile RA PDOs and observed highly diminished perivascular TGFβ isoform expression, with de-repression of TGFβ receptors, in DAPT treated organoids (**Fig. 7i, Extended Data Fig. 13d**) as well as diminished COMP expression in TGFβi treated organoids **Fig. 7j**). Collectively, our data supports active maintenance of perivascular-compartmentalized TGFβ isoform and receptor regulation in synovial tissue that can be reversed



with Notch inhibition, and additionally the ability to abrogate the fibrogenic gene expression programs in synovial tissue with TGFβ inhibition.

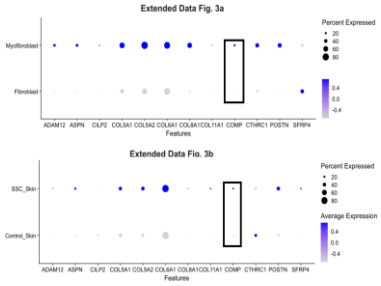
R1C4. "*There are numerous mentions of the immune landscape in your samples. Of note in Figure 7, your non-remission patient sample has a striking immune phenotype prior to treatment. The data presented in Figure 7b makes it difficult to assess if and how immune cells are important in the non-remission phenotype. Is there correlation between immune cell number prior to treatment and COMPhi fibroblasts?*"



We thank the reviewer for their questions regarding the relationship between immune cell landscape, remission status and *COMP*+ cell abundance. To address this, we extended our niche analysis to post-treatment biopsies (**Fig. 7b**). We agree that one of the non-remission patients had high immune niche abundance that was sustained post-treatment; however, there were no significant differences in immune niche composition pre- or post-treatment between remission groups (**Extended Data Fig. 2f,11f**). Additionally, we observed a generally negative correlation between *COMP*+ fibroblast abundance and immune cell percentage. Clinically (please see updated **Supplementary Table 1, R3C1**), swollen joint count and ESR did not differ between groups at baseline or after 6 months, further suggesting that inflammation status is not the main driver of non-remission. In contrast, non-inflammatory disease activity components including tender joint count and VAS pain drove non-remission in our cohort.

R1C5. "*Is COMP expressed in other disease indications? Including it in the IPF and SSc analysis in Extended Fig2 may be interesting*"

We appreciate the reviewer's suggestion to show *COMP* expression in fibrotic disease. We have included dotplots that highlight increased *COMP* expression in IPF myofibroblast populations and SSc-derived fibroblast populations (**Extended Data** Fig. **3a,b**)

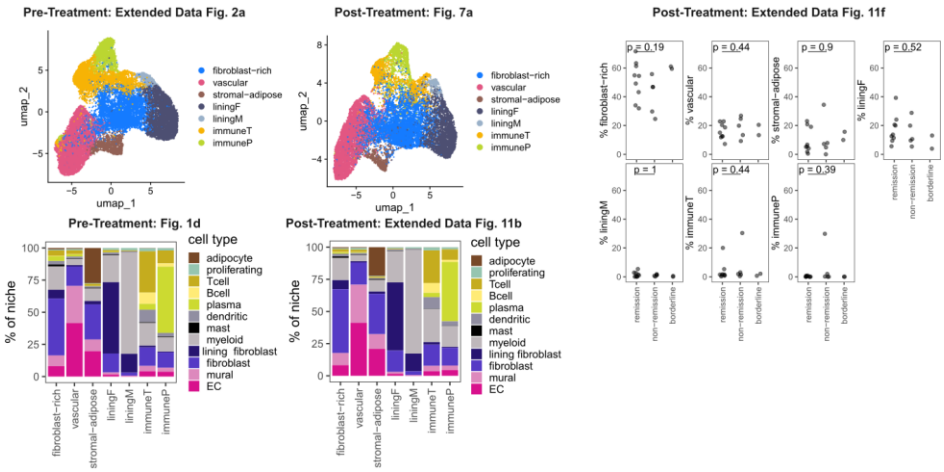


Reviewer #2

R2C1 "*Most importantly, the authors fail to provide a similar niche and cell analysis in post-treatment spatial transcriptomics. This exclusion questions the premise of delineation of the niches and the cell attributions in the treatment-naïve samples. Without a comparative section on pre- and post-treatment spatial analysis, the merit of the conclusions may not hold…Surprisingly, there are no statistically significant differences in the niche composition between baseline and remission status. This warrants examination and explanation, and this questions the niche*

assignments. This is another reason for the examination of post-treatment spatial transcriptomics in detail."

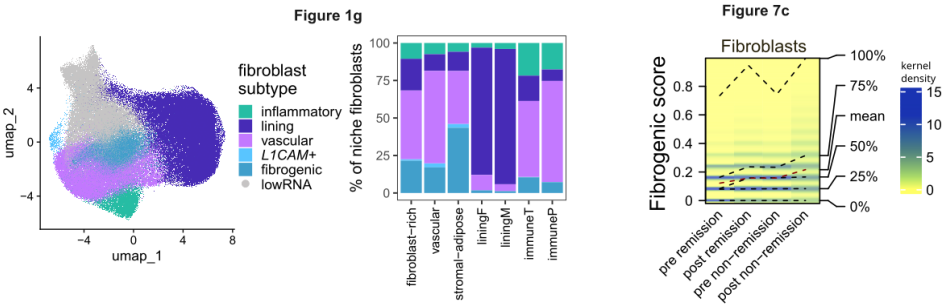
We agree with the reviewer's suggestion that a comparative pre-and post-treatment niche analysis is essential for both verifying the validity of niche delineation and providing additional insights towards changes in spatial composition post-treatment. To address this, we performed comprehensive niche analysis of post-treatment biopsies. First, we compared the consistency of niche delineation in pre- (**Fig. 1d, Extended Data Fig. 2a**) versus post-treatment (**Fig. 7a, Extended Data Fig. 11b**) biopsies by analyzing cellular composition in both contexts and observed very high concordance with how niches were defined across timepoints, suggesting the stability of niche calling with the methodology we used[1]. Of note, we have updated cellular composition plots to distinguish lining fibroblasts from non-lining fibroblasts. Additionally, we updated **Fig. 7b** (see **R1C4**) to include plots that show changes in niche composition, rather than changes in cellular composition, pre- and post-treatment. Consistent with cell-level depletion of lymphocytes and plasma cells post-treatment, ImmuneP and ImmuneT niches were depleted, further supporting the validity of our niche calling. We also analyzed changes in niche composition post-treatment by remission status and as observed in baseline biopsies, we did not detect major differences across groups. We did, however, observe significant post-treatment expansion of the fibroblast-rich niche across patients suggesting that niche composition is affected more by treatment than by remission outcome. Overall, we find that our niche calling is consistent with known synovial architecture and post-treatment changes in niche-composition are consistent with previously reported findings[2].



R2C2 "*The identification of fibroblast phenotypes associated with the non-response is interesting, and a higher population of fibrogenic fibroblasts was found. It would be important to ask what other cell types were associated with the fibroblast niches and specifically if the fibrogenic fibroblasts were present in the lining regions. The authors define a 10-gene fibrogenic signature (based on prior literature). What is not clear from the paper is whether this signature is present in regions in both responders and non-responders. It would be valuable to project these on the niche UMAP to see if these are present in pre-treatment and post-treatment (responders vs non-responders).*"

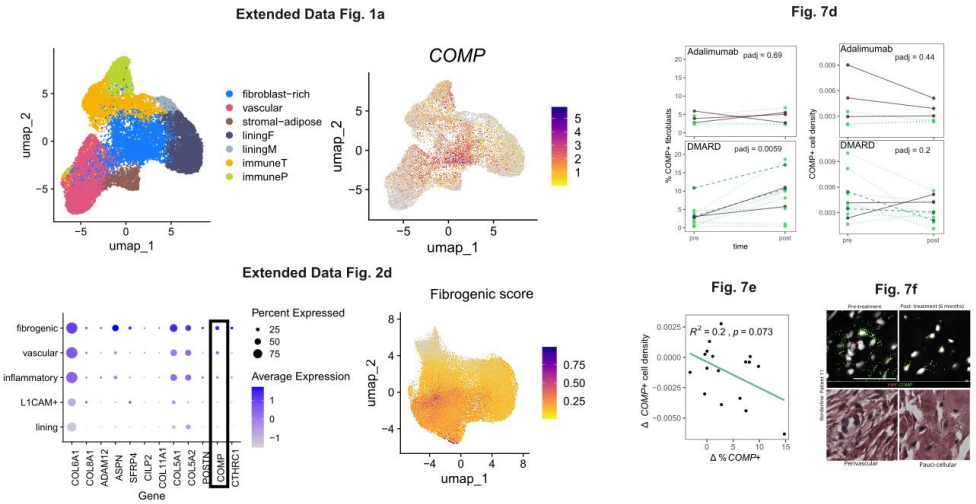
We thank the reviewer for their comment and agree that more analysis regarding fibrogenic fibroblast localization in synovium and the expression of the fibrogenic signature in relation to response and treatment status are highly important. To address this, we included a plot to highlight the distribution of fibroblast subtypes across different niches (**Fig. 1g**). We found that the fibrogenic subset was most prominent in the vascular, fibroblast-rich and stromal-adipose niches rather than in lining niches. Additionally, we included analyses in which we quantified

fibroblast fibrogenic signatures across remission status, finding that while it is present in both response groups, it was significantly elevated among non-remission patients (**Fig. 1h,** comment **R1C1**). We also identified the vascular niche as the site of differentially elevated fibrogenic signature scores in non-remission (**Fig1j,k**, comment **R1C1**). Lastly, we added a density plot (**Fig. 7c**) that highlights fibrogenic fibroblast signature scores based on treatment and response status. We found that fibrogenic scores increased across remission groups, but that the post-treatment increases in fibrogenic signature was more prominent in non-remission patients.



R2C2. "*The entire discussion on COMPhi appears to be distinct from the 16-patient study that forms the core of this research. The authors use a completely different dataset, the AMP RA/SLE single cell RNA seq, subclustered their single cell fibroblast expression, and pulled out COMP and POSTN expression. They have not examined in detail the expression of POSTN and COMP in the pre- and post-treatment niches. How statistically significant is the expression difference in the responder vs non-responder in their dataset? clearly, their power is low, but it would be important. It does not make sense to compare a spatial data set with a single-cell data set with spatial fibroblasts in different niches compared to a scRNA fibroblast sub clustering. The authors state (lines 213-217) that COMP but not DKK3, FMOD, or POSTN were elevated in treatment-naïve biopsies. How about post-treatment biopsies? This "big claim" on COMPhi hinges on this questionable hypothesis.***"**

We appreciate the reviewer's comments and agree with them that it is important to use our spatial transcriptomic dataset, rather than alternate datasets, as the basis for analyzing *COMP* expression patterns. To address these comments, we have substantially re-organized our analysis of fibrogenic fibroblasts in RA and performed deeper analysis of *COMP* expression in our cohort. While we had previously derived a 10-gene fibrogenic signature from fibrotic disease datasets, we have now defined the fibrogenic signature from our RA spatial data instead, using the top 12 marker genes in the fibrogenic fibroblast cluster (**Extended Data Fig. 2d**) of which *COMP* was among the most highly expressed genes and also highly expressed in the fibroblast-rich niche (**Extended Data Fig. 1a**). We then



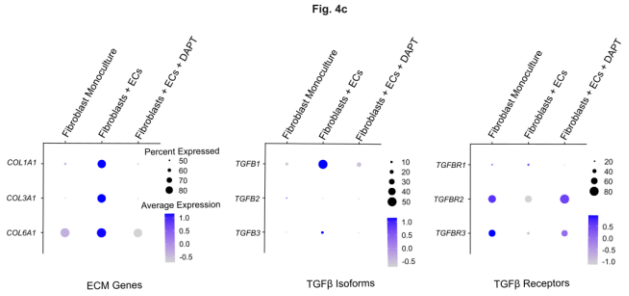
applied the spatially derived fibrogenic signature to AMP single cell RNA sequencing data to gain further insight into the heterogeneity of fibrogenic fibroblasts, finding that *COMP* marked a subset of fibrogenic fibroblasts that was particularly non-response associated (**Fig.** 1i, Extended **data Fig. 4**). Further, we have thoroughly quantified *COMP* expression in both pre- and post-treatment biopsies finding that 1) baseline *COMP* transcript expression is higher in non-remission (**Fig. 1i, R1C2**) and 2) that the *COMP*+ cell population expands significantly across remission groups post-treatment (**Fig. 7d**). Spatially, we found that the expanded population of *COMP*+ fibroblasts post-treatment accumulates within pauci-cellular fibroblast-rich niches (**Fig. 7e,f**). Clinically, we found that baseline *COMP* transcript levels in our cohort correlate with lesser improvements in tender joint count, among non-remission, and higher DAS28-ESR at 12 months (**Extended data Fig. 4g, R1C2**). Collectively, our analysis of spatial data revealed *COMP* expression as a marker of non-remission.

R2C3 "*To prove their hypothesis that TGFβ role in fibroblasts driven by notch signaling, the authors use a co-culture of HUVEC and synovial fibroblasts. There are two fundamental issues with this approach. Most importantly, they use synovial fibroblasts when their entire argument in their spatial analysis indicated vascular fibroblasts. Second, this "artificial system" is devoid of the entire immune environment and the nearby niches in the patient tissue, so any conclusions from here may have no relevance to the in situ system. This appears highly artificial. Why not investigate the spatial distribution of the genes in the different niches, including the fibroblast niches? All the validation experiments for the notch patterning role in fibrogenic program are performed on the co-culture model. Such experiments cannot be done on tissues, but the entire set of experiments is biologically questionable, and the statement that the in vitro and in vivo results suggest that Notch tunes TGF sensitivity is speculative.*"

We thank the reviewer for their comment and agree with them that the use of additional model systems is essential to connect the mechanistic findings from *in vitro* co-culture to an *in vivo* relevant context. To address this comment, we developed a novel RA patient-derived synovial organoid (PDO) approach that maintains synovial tissue architecture *ex vivo* (please see comment **R1C3, Fig. 7g**) and spatially profiled this system. We found that 1) inhibition of Notch or TGFβ signaling substantially reduces fibrogenic protein production from synovial organoids (**Fig. 7h**). 2) Notch inhibition reduces perivascular TGFβ isoform expression and leads to de-repression of TGFβ receptor expression (**Fig. 7i, Extended Data Fig. 13d**). 3) That TGFβ inhibition significantly diminishes *COMP* expression (**Fig. 7j**). Collectively, the observations from our *in vitro* co-culture data were highly concordant with what we observed by directly manipulating Notch and TGFβ signaling in synovial tissue organoids, thereby supporting that Notch-mediated tuning of TGFβ is conserved across model systems.

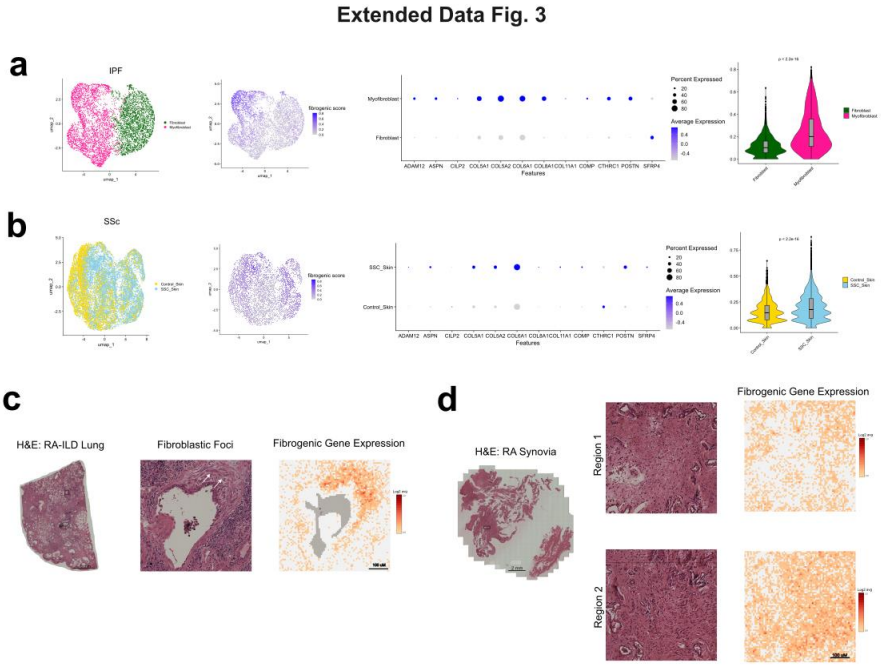
R2C4 "*In the patterning of fibroblast TGFβ responsiveness, the authors use yet another model system – Matrigel-embedded 3D organoids of fibroblasts and ECs with and without gamma secretase inhibitor. The authors show that TGFβ1 and TGFβ3 are induced with fibrogenic genes, but the TGFβ receptors are suppressed. The heat map in Fig. 4b is ambiguous and does not appear to corroborate this finding.*"

We thank the reviewer for their comment and have used a dotplot instead of a heatmap to emphasize the regulation of TGFβ isoform and receptor expression across conditions.



R2C5 "*The 10-gene signatures obtained from prior studies on SSc and IPF datasets are associated with scleroderma-associated and myofibroblasts. It is not clear if the cell types in the treatment-naïve or post-treatment spatial deconvolution involved myofibroblasts. These are distinct from fibrogenic fibroblasts described in this study.*"

We thank the reviewer for their suggestion to clarify the connection between RA fibrogenic fibroblasts and myofibroblasts in fibrotic diseases. To address this question, we have performed new spatial transcriptomic experiments utilizing a whole-transcriptome technique (Visium HD, 10X Genomics) of lung tissue from an RA-ILD patient exhibiting usual interstitial pneumonia. As described in response to **RC2C2**, we modified our analysis by first defining the fibrogenic fibroblast signature from our RA spatial transcriptomic data (**Extended data Fig. 2d**), and then we mapped the RA-defined fibrogenic signature to IPF and SSc datasets, finding that the RA-defined signature was strongly expressed in the "myofibroblast" annotated IPF cluster (**Extended data Fig. 3a,b)**. Additionally, we spatially mapped the signature to RA-ILD tissue and found that the signature was specifically enriched in bona fide fibroblastic foci as determined by histological assessment **(Extended data Fig. 3c,d**). Collectively, we believe that the RA fibrogenic fibroblast population is analogous to myofibroblast populations in fibrotic disease.



R2C6 "*The authors find that in treatment-naïve biopsies of patients who fail to achieve remission (how many patients?), COMP but not POSTN or FMOD are elevated. COMP expression is triggered by SOX9, which is downstream from TGFβ. Was there evidence of SOX9 expression? The story of the antagonistic relationship between POSTN and COMP is not mechanistically observed since while they are both downstream from Notch-TGFβ, they are independently*

regulated (not anticorrelated in cartilage expression. Are the authors proposing a new mechanism?"

We thank the reviewer for their questions regarding the regulation of *COMP* expression. We are aware of prior literature that describes the role of *COMP* and SOX9 in chondrogenesis[3,4]. In spatial transcriptomic profiling of our synovial biopsies, we did not detect any chondrocyte-like populations because these tissues were obtained by sampling only the synovial membrane, rather than the entire joint. To address the reviewers' comment regarding SOX9, we examined whether there was co-expression of *SOX9* and *COMP* in synovial fibroblasts and found that *SOX9* is co-expressed in a very small subset of *COMP*-expressing fibroblasts. The role of *SOX9* in regulating *COMP* expression in fibroblasts will require further investigation out of the scope of this current study.

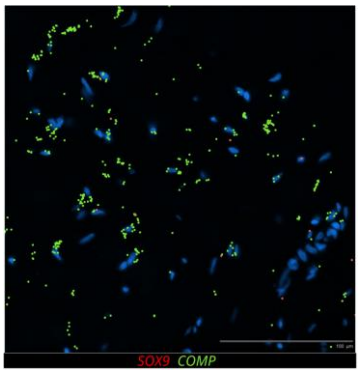


Figure 4j

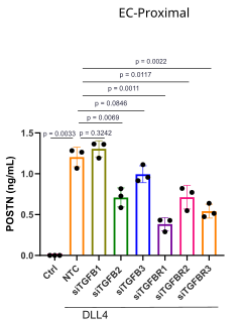
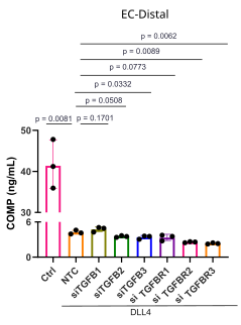
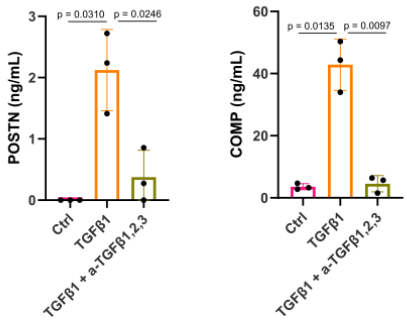


Figure 4k



Extended data Fig. 8f

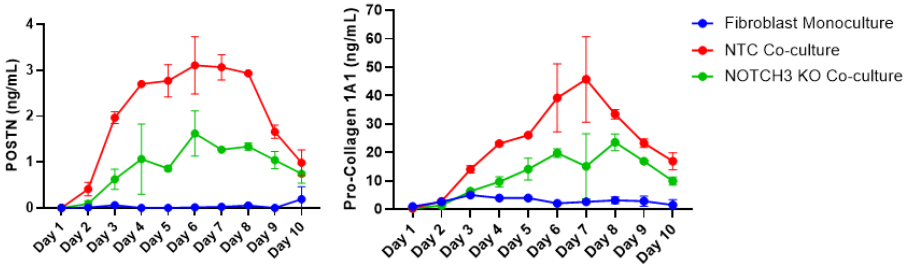


Regarding the reciprocal regulation of *POSTN* and *COMP* expression, we propose that *POSTN*, which is significantly induced by both Notch and TGFβ stimulations (**Fig. 4j, Extended data Fig. 8f**), is controlled by the promoters of both signaling pathways. In contrast, *COMP* is highly induced by TGFβ stimulation but suppressed by Notch signaling (**Fig. 4h**, **Extended data Fig. 8f)** in a manner that can be rescued by TGFβRIII overexpression (**Fig. 5i**). Therefore, the novel mechanism we propose in the manuscript is that *COMP* expression is driven by TGFβ signaling but can be suppressed in the context of Notch-mediated suppression of TGFβ receptors.

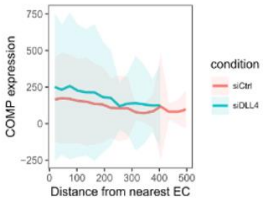
R2C7 "*In the discussions on notch and TGFβ in co-culture experiments, there is significant discussion on the isoforms of TGFβ, while there is no discussion on notch isoforms. Based on prior literature, Notch3 is the isoform that is upstream of COMP and POSTN expression. Did the authors explore which Notch isoform is expressed and which isoform controls the TGFβ pathway? Also, the discussion on the TGFb1, 2, 3 is confusing. It would be good to summarize this in a pathway schematic. Unless this reviewer is mistaken, DLL4, which induces TGFβ1 is a ligand for Notch 1 and Notch 2 but not Notch 3. Does this mean Notch 3 is not involved in the fibrogenic process in non-remission? This is confusing. Also, the Notch signaling pathway can regulate COMP protein production, according to studies on cartilage development. Specifically, the Notch pathway influences the differentiation of mesenchymal progenitor cells (MPCs), which participate in cartilage formation, and thus can impact COMP protein expression. However, are the fibrogenic fibroblasts these mesenchymal cells? Some clear discussion on this is merited.*"

We thank the reviewer for seeking clarification on relevant Notch and TGFβ isoforms. Regarding TGFβ isoforms, we have added a citation (reference 37) that comprehensively describes their roles. Briefly all three isoforms signal identically; however, TGFβ2 and 3 have intrinsic activity that

does not require the cleavage of a latency domain as is necessary for TGFβ1 activity. Regarding Notch isoforms, our previous work[5] and other studies[6,7] have established that Notch3 can bind DLL4, and further that Notch3 is the relevant receptor isoform expressed on perivascular stromal populations[8,9]. To test the requirement of Notch3 in mediating fibrogenic protein production, we performed 3D co-cultures of HUVECs with fibroblasts that were CRISPR-modified with non-targeted control (NTC) or *NOTCH3* guide RNAs and measured supernatant protein production of Pro-Collagen I and POSTN. We found that co-cultures with *NOTCH3* CRISPR knockouts had significantly diminished production of both fibrogenic proteins, supporting the role of Notch3 in mediating the fibrogenic signaling. Regarding the origin of fibrogenic fibroblasts, we believe, based on previous literature [10], that they arise from mesenchymal stem/progenitor cells. However, as described in **R2C6**, we found that Notch signaling suppresses *COMP* expression via its suppression of fibroblast TGFβ receptors; we therefore believe that Notch signaling alone is unable to induce *COMP* expression in fibroblasts, though we acknowledge its potential role in driving chondrocyte *COMP* expression.



R2C8 "*Does the discussion on the distance of COMPhi cell being higher in DLL4 knockdown refer to EC proximity? Figs 6e and 6f do not seem to clearly state this point.*"



We thank the reviewer for their question and have modified the text to clarify: "we observed that average *COMP* expression was higher with DDL4 knockdown compared to control and that the average distance of *COMP*hi cells to ECs increased with DLL4 knockdown (p = 0.00367) (**Fig. 6e-f**)". We would like to highlight that we included a line plot (**Fig. 6f**) that shows *COMP* expression levels in relation to endothelial cell distance. As can be observed from the presented data, average *COMP* expression in the DLL4 knockdown condition is higher in both proximal and distal regions, up to 400 µm.

R2C9 "*In the discussion of COMPhi regions, it is stated they become pauci-cellular (Fig 7e, g). Does this refer to pauci-immune or pauci-fibroblast? Again, connecting this observation to niches in post-treatment responsive and nonresponsive patients would help.*"

We thank the reviewer for their suggestion to place *COMP*hi fibroblast localization in the context of post-treatment niche analysis. Accordingly, we have added post-treatment niche analysis (**Fig. 7b, R1C4**), showing that there is substantial contraction of immune compartments with concomitant expansion of fibroblast-rich niches where we found that the *COMP*hi fibroblasts accumulate (**Fig. 7d-f, R2C2**). Therefore, the areas where the fibroblasts accumulate are both pauci-immune and pauci-fibroblast which we have updated the text to clarify.

Reviewer #3

R3C1 "*Clinical: the spatial transcriptomic data are derived from a small cohort of 17 patients in the 396.10 study classified according to DAS28-ESR remission status (supplementary table 1). A*

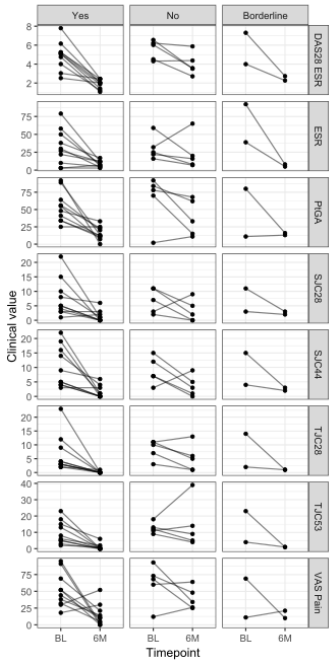
more complete description of the clinical outcomes would aid in interpretation of the biopsy data. Given that the DAS score is a composite measure of swollen and tender joint counts, inflammatory biomarkers, and a subjective report of pain/disability, it would be helpful to understand which of those components drove remission or non-remission in this study. Of note, at baseline, the "non-remission" patients appear to have higher DAS scores than the "remission" patients, but is this because they have more active joints, more inflammatory biomarkers, more pain, or some combination thereof? Remission is defined as a 6-month DAS28-ESR score <2.6 but it's not clear whether the "non-remission" patients experienced any decrease in DAS score on treatment (but remained >2.6) or had different patterns of response from the "remission" patients. For example, in the PEAC cohort (figure 1f), the definition of "non-response" is fairly strict: DAS28-CRP > 5.1 and improvement < 0.6. In 396.10, did "non-remission" patients with a decrease in inflammatory infiltrates in their synovial biopsies exhibit meaningful changes in joint counts or ESR/CRP but not pain? In the Discussion section (lines 523-527), the authors lean heavily on "non-inflammatory pain" as being a driver of incomplete response to anti-inflammatory treatment; is this concept applicable specifically to the patients examined in the 396.10 study? What evidence is present in this study that "non-inflammatory pain" is indeed linked to fibrogenic fibroblast populations?"

We appreciate the reviewer's recommendation to further analyze DAS components to identify which clinical features are relevant to remission status pre- and post-treatment. Accordingly, we have updated **Supplementary Table 1** to include the comparison of DAS components between remission and non-remission patient groups at baseline and post-treatment. While there was indeed a trend that non-remission (n=5) patients at baseline had higher baseline DAS28-ESR scores than remission patients (n=10), this was not statistically significant (p=0.4396), nor was there any significant difference observed in any individual component of the DAS28-ESR (TJC53, p = 0.0.3570; SJC44, p=0.9022; patient global assessment (PtGA), p = 0.3263; ESR, p = 1). At 6 months post-treatment, tender joint count and VAS pain (TJC28, p = 0.0011; TJC53, p = 0.0037, VAS Pain p = 0.0233) were the components that drove non-remission rather than inflammation-related components such as swollen joint count (SJC44, p = 0.2273), which is consistent with the substantial depletion of immune niches observed across groups. Additionally, we included a line plot showing the differences pre- and post-treatment; while DAS components did improve across remission groups, pain-related DAS components were the drivers of sustained disease activity in non-remission at 6 months. Further, we analyzed *COMP* transcript expression in relation to DAS and its components (**Extended Data Fig. 4g**, **R1C2**) and found that it correlated with less improvement in tender joint count among non-remission patients (deltaTJC28 at 6 months and 12 months) and had a positive association with DAS28-ESR at 12 months across the remission and non-remission groups. Altogether, analysis of clinical data from our cohort supports non-inflammatory pain as the sustained driver of disease activity in non-remission and its link with fibrogenic signaling.

	Total	remission	non-remission	borderline	remision vs non-remission	
	(N=17)	(N=10)	(N=5)	(N=2)	p-value	significance
Sex						
Female	8 (47%)	4 (40%)	3 (60%)	1 (50%)	0.6084	No
Male	9 (53%)	6 (60%)	2 (40%)	1 (50%)		
Treatment						
DMARD	11 (65%)	7 (70%)	2 (40%)	2 (100%)	0.3287	No
Adalimumab	6 (35%)	3 (30%)	3 (60%)	0 (0%)		

	Mean(SD)	Mean(SD)	Mean(SD)	Mean(SD)		
Age (y)	47.35 (17.45)	50.50 (22.18)	43.20 (3.11)	42.00 (12.73)	0.5941	No
DAS28-ESR (baseline)	5.17 (1.41)	4.89 (1.50)	5.52 (1.04)	5.65 (2.34)	0.4396	No
DAS28-CRP (baseline)	4.69 (1.28)	4.52 (1.34)	5.01 (1.04)	4.77 (2.20)	0.4396	No
TJC53 (baseline)	11.06 (6.68)	9.80 (7.07)	12.60 (3.36)	13.50 (13.44)	0.3570	No
SJC44 (baseline)	9.65 (6.14)	10.10 (7.06)	8.80 (4.71)	9.50 (7.78)	0.9022	No
TJC28 (baseline)	7.24 (5.76)	6.50 (6.65)	8.40 (3.44)	8.00 (8.49)	0.2647	No
SJC28 (baseline)	7.29 (5.52)	7.60 (6.50)	6.80 (4.27)	7.00 (5.66)	1.0000	No
ESR (baseline)	35.59 (25.13)	32.00 (24.81)	30.80 (16.78)	65.50 (37.48)	1.0000	No
PtGA (baseline)	56.18 (28.44)	53.80 (22.66)	65.20 (36.24)	45.50 (48.79)	0.3263	No
VAS Pain (baseline)	53.35 (27.32)	52.10 (25.73)	61.20 (30.08)	40.00 (41.01)	0.5002	No
DAS28-ESR (6 months)	2.59 (1.20)	1.90 (0.50)	4.01 (1.20)	2.49 (0.33)	0.0007	Yes
DAS28-CRP (6 months)	2.41 (1.06)	1.74 (0.37)	3.74 (0.94)	2.45 (0.18)	0.0007	Yes
TJC53 (6 months)	4.82 (9.64)	0.90 (1.91)	14.20 (14.41)	1.00 (0.00)	0.0037	Yes
SJC44 (6 months)	2.24 (2.59)	1.50 (2.12)	3.60 (3.58)	2.50 (0.71)	0.2273	No
TJC28 (6 months)	1.71 (3.41)	0.10 (0.32)	5.20 (4.92)	1.00 (0.00)	0.0011	No
SJC28 (6 months)	2.00 (2.57)	1.30 (1.95)	3.20 (3.83)	2.50 (0.71)	0.4377	No
ESR (6 months)	12.65 (14.33)	8.60 (4.48)	23.20 (23.99)	6.50 (2.12)	0.0964	No
PtGA (6 months)	21.94 (18.36)	15.50 (9.70)	37.80 (26.26)	14.50 (2.12)	0.1107	No
VAS Pain (6 months)	22.00 (18.82)	14.60 (16.21)	39.40 (16.55)	15.50 (7.78)	0.0233	Yes
delta DAS28-ESR	-2.58 (1.56)	-2.99 (1.49)	-1.51 (1.34)	-3.16 (2.02)	0.0400	Yes
delta DAS28-CRP	-2.28 (1.55)	-2.78 (1.45)	-1.27 (1.28)	-2.31 (2.38)	0.0553	No
delta TJC53	-6.24 (10.05)	-8.90 (7.08)	1.60 (11.48)	-12.50 (13.44)	0.1097	No
delta SJC44	-7.41 (6.77)	-8.60 (7.29)	-5.20 (6.46)	-7.00 (7.07)	1.0000	No
delta TJC28	-5.53 (5.94)	-6.40 (6.74)	-3.20 (3.35)	-7.00 (8.49)	0.7108	No
delta SJC28	-5.29 (6.23)	-6.30 (6.91)	-3.60 (5.94)	-4.50 (4.95)	0.9510	No
delta ESR	-22.94 (28.20)	-23.40 (23.49)	-7.60 (26.11)	-59.00 (35.36)	0.4396	No
delta PtGA	-34.24 (28.55)	-38.30 (27.94)	-27.40 (29.23)	-31.00 (46.67)	0.5399	No
delta VAS Pain	-31.35 (34.96)	-37.50 (37.21)	-21.80 (30.78)	-24.50 (48.79)	0.5941	No

Extended Data Fig. 11g



R3C2 "*More detail on biopsies is needed. How do we know the associations between spatial transcriptomic assessments and clinical outcomes over time are based on "representative" biopsies of "representative" joints and not simply due to sampling artifacts? Which joints were biopsied, were they selected via ultrasound assessment for active synovitis, and were the same or different joints assessed at baseline and 6 months?***"**

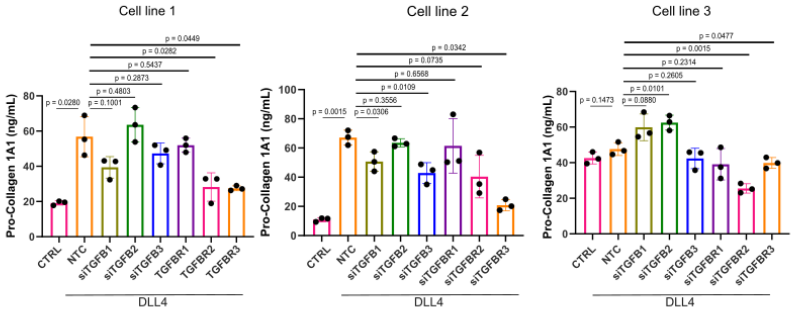
We appreciate the reviewer's recommendation to improve our description of the biopsy method to ensure representative sampling. For RA synovial biopsies, we used a small-bore arthroscopic method for greater yield and a larger area of synovial tissue per fragment biopsied which gave us the full range of cellular subsets from the lining all the way to the deep sublining, as opposed to the smaller fragments from ultrasound-guided biopsies. All biopsies were undertaken in an active joint (typically the knee joint) and were performed under direct vision (with documentation of degree/ extent/ severity of synovitis). The same joint was biopsied at baseline and 6 months. We have updated the methods section to provide more detail regarding the biopsy method.

R3C3 "*This study raises many questions about TGFbeta signaling: via several in vitro mechanistic models, the authors conclude that the main effect of endothelial-derived Notch signals on fibroblasts is the spatial regulation of TGFBR3 expression (lines 435-440). While there are differential effects on TGFb isoform expression, it appears that TGFb1 and b3 are more significantly affected than TGFb2, and the authors suggest that TGFb3 may be the isoform responsible for the COMPhi fibroblast population expansion. However, relatively scant attention is paid to the previously described roles for TGFb receptor subunits and selectivity for isoforms: whereas TGFb1 and b3 bind strongly to TGFBR2 and recruit TGFBR1 (ALK5) to induce intracellular signaling, TGFb2 has lower affinity for TGFBR1 and 2 (e.g., PMID 20207738). Many studies have suggested that TGFBR3 specifically acts to facilitate TGFb2's ability to bind to TGFBR1/2 complexes and direct signaling, whereas its effects on other TGFb isoforms are minimal. Is it therefore possible that TGFb2, rather than TGFb3, is the potential therapeutic target? In figure 4j, siRNA knockdown of each TGFb isoform individually (including TGFb2) in*

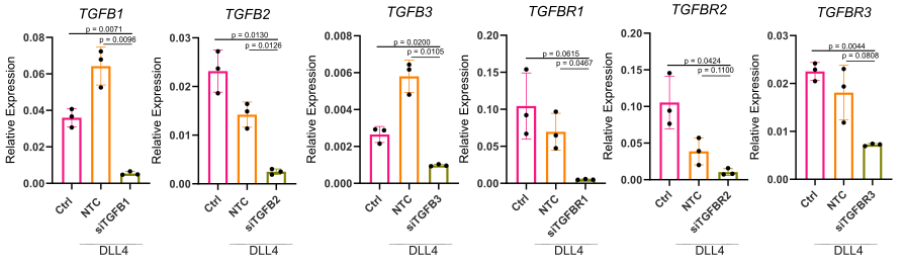
DLL4-treated cells shows comparable modest reductions in pro-Col1A1 production despite no differential effect of DLL4 +/- DAPT treatment on TGFb2 expression (figure 4e), but less reduction than knocking down either TGFbR2 or R3. Why is there no effect when TGFbR1 is knocked down (given that ALK5 is required for SMAD activation and signaling, e.g., in figure 4i, where an ALK5 inhibitor is used)? While there are data to show the efficiency of DLL4 knockdown (extended data figure 5), there is not similar confirmation of the knockdown efficiency of the other targets, which may be a simple explanation for the divergent results. Given the tight regulation of TGFb positioning by milieu molecules (e.g., LTBRs, LRRCs) and activation of latent TGFb (e.g., by integrin binding or LAP cleavage), do the authors believe that TGFb signaling is autocrine, paracrine (i.e., dependent on close cell-cell contact), or via broader diffusion of soluble active TGFb? This in turn may be highly dependent on which isoform is being considered: whereas latent TGFb1 has a very high threshold for activation, latent TGFb2 and TGFb3 have intrinsic activity."

We appreciate the reviewer's comments to more thoroughly explore TGFβ isoforms, receptors, and signaling, and their recommendations have helped us to substantially improve the manuscript with additional experimental data. Regarding TGFβ isoforms, we have added the recommended reference [11] that highlights the intrinsic ability of TGFβ2 and TGFβ3 to signal without cleavage of their latency domains We have also included a reference to a recent study that highlights the ability of TGFβRIII to strongly bind TGFβ3 in addition to TGFβ2[12]. While we believe that binding of TGFβ2 to TGFBRIII is crucial for its activity in many contexts [13,14], we propose that the high expression of TGFβ3, relative to TGFβ2, in perivascular stromal populations and fibrogenic fibroblast populations implicates TGFβ3 as a possible mediator of fibrogenic signaling *in vivo*[15]. To address the role of relevant TGFβ isoforms and receptors *in vitro* we have now performed new experiments and included a comprehensive siRNA knockdown data and evaluated whether Notch-mediated TGFβ signaling occurred in a soluble manner. First, we expanded the number of synovial fibroblast cell lines (n > 6 and showed data from 3 representative cell lines in the manuscript) in which we conducted siRNA knockdown, including combinatorial knockdown, of receptors and isoforms and performed RT-qPCR to validate the knockdown efficiency across cell lines (**Fig. 4i**, **Extended data Fig. 8g**). Despite high knockdown efficiency, we found variable results across RA fibroblast lines regarding the ability of TGFβ isoforms knockdown (single or combinatorial) to abrogate pro-collagen type I expression, concluding that they likely drive signaling in a redundant manner i*n vitro* and variably across RA patient-derived fibroblasts (**Extended data Fig. 8h**). Notably, we have observed that *TGFB1* is much more highly expressed in synovial fibroblast cell lines (>5-fold on average)

Figure 4i

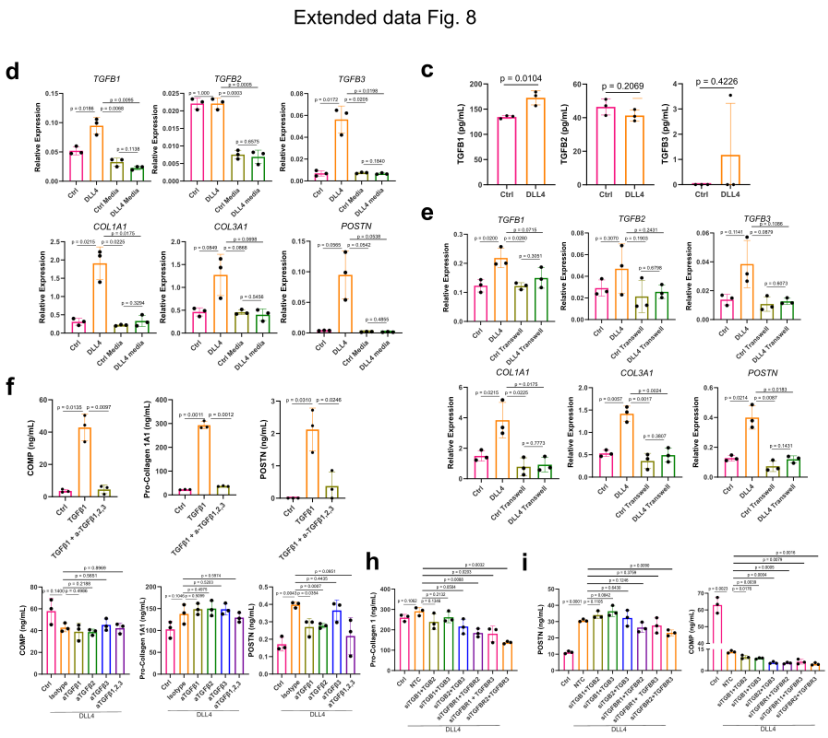


Extended Data Fig. 8g



compared to *TFGB3*, but the knockdown of *TGFB1* did not lead to significantly stronger abrogation of pro-collagen type I expression compared to *TGFB3* knockdown. Additionally, knockdown of *TGFB2* did not strongly reduce Notch-mediated fibrogenic protein production; therefore, it is difficult for us to conclude, based on *in vitro* data, if a specific isoform is dominant over the other isoforms in driving TGFβ signaling. We have modified our results and conclusion to reflect this in the revised manuscript. Regarding TGFβ receptors, we found that knockdown of *TGFBR2* and *TGFBR3* (single and combinatorial) led to the most substantial reductions in pro-collagen production compared to knockdown of *TGFBR1* despite high siRNA efficiency across cell lines (**Fig. 4i, Extended data Fig. 8g**). We therefore believe that *TGFBR2* and *TGFBR3* expression levels more strongly gate TGFβ responsiveness than *TGFBR1* expression levels, though we acknowledge the necessity of TGFβRI in mediating TGFβ signaling. We also included additional data in the manuscript that reveals similar isoform and receptor-specific knockdown effects on the production of COMP and POSTN (**Fig. 4j,k** in **R2C6**, **Extended Data Fig. 8i**)

We also tested whether Notch-mediated TGFβ signaling occurred in a soluble manner with 4 additional experiments. First, we analyzed supernatant levels of soluble TGFβ isoforms on DLL4-stimulated fibroblasts, finding that all isoforms were detected at low levels (< 0.2 ng/mL) compared to levels (> 1 ng/mL) that are required to elicit comparable induction of fibrogenic protein expression (**Extended Data Fig. 8c**). Next, we treated fibroblasts with conditioned media from DLL4-stimulated fibroblasts that showed strong TGFβ isoform and fibrogenic gene expression and observed non-significant differences between fibroblasts treated with conditioned media from control versus DLL4-stimulated fibroblasts (**Extended Data Fig. 8d**). We then cultured fibroblasts on transwell inserts above control or DLL4-stimulated fibroblasts that similarly showed strong TGFβ isoform and fibrogenic gene expression and observed nonsignificant differences between fibroblasts cultured above uncoated versus DLL4-coated wells that were seeded with fibroblasts (**Extended Data Fig. 8e**). Lastly, we used antibody blockade of individual and pan-TGFβ isoforms on DLL4-stimulated cells, finding that they did not significantly influence fibrogenic protein production, despite being highly effective at neutralizing the effects of recombinant soluble TGFβ on fibrogenic protein production (**Extended Data Fig. 8f**). Altogether, we conclude that Notch-mediated TGFβ signaling likely occurs in a contact-dependent autocrine or paracrine manner rather than in a soluble manner, which we have now described in the manuscript.

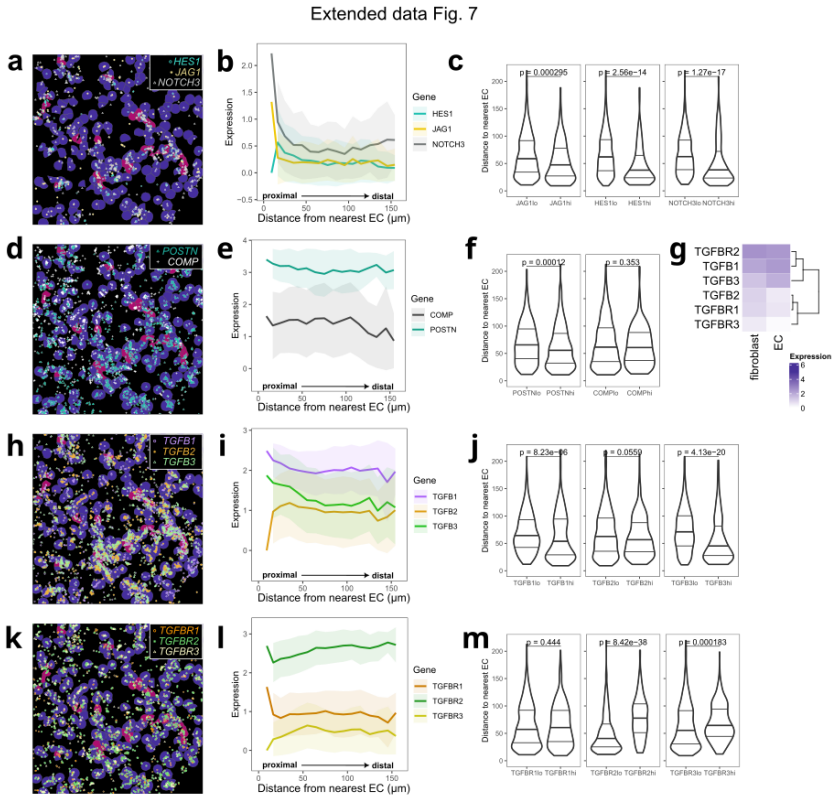


R3C4 "There *are several different in vitro models employed in figures 3-6 but it is confusing why certain models were used for certain purposes. It appears that the organoid model data in figure 4a-d are derived from a dataset in a previous publication from this group (ref. 9) but the "new" experiments in this study are in a*

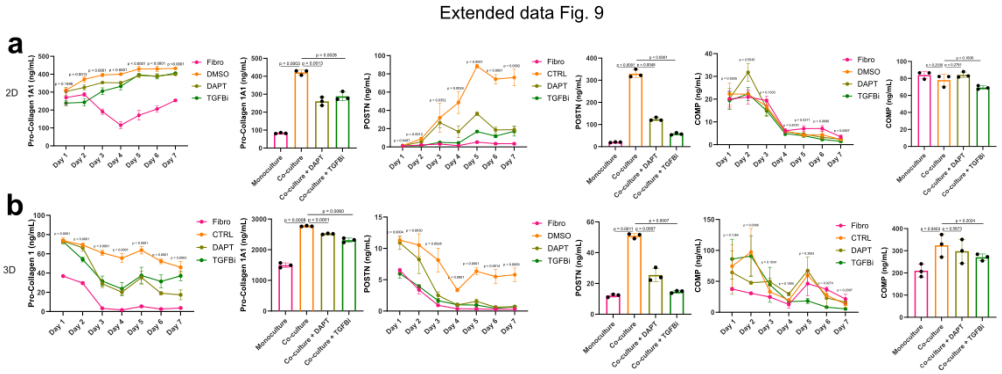
simpler 2-D co-culture system with HUVECs and synovial fibroblasts. I have no issues with recycling data from a prior study but am curious whether the patterns of gene expression and cellular positioning identified in the 3-D organoids and the 2-D culture system align with each other. In particular, the role of milieu molecules and ECM in positioning and activating TGFb isoforms (see comment #3, above) may be different across these experimental systems and/or differentially relevant to an in vivo setting.

We thank the reviewer for encouraging us to more deeply examine fibrogenic signaling in a 3-D context, which may be more *in vivo* relevant. To address these questions, we have performed new experiments and added a substantial amount of experimental data from 3-D culture experiments, including spatial transcriptomic profiling of 3-D co-cultures to quantify patterns of gene expression (**Extended Data Fig. 7**). Briefly, we resuspended a 1:1 mixture of endothelial cells and fibroblasts (100,000 cells each) in matrigel to form 3D spheroids that were cultured for 14 days, then fixed the spheroids, embedded them in FFPE blocks and sectioned them onto Xenium slides for spatial transcriptomic profiling. We observed high concordance in patterning of gene expression between the 2-D and 3-D cocultures observing, in the EC-proximal compartment, 1) high Notch activation (**Extended data Fig. 7a-c),** 2) high *POSTN* expression (**Extended data Fig. 7d-f**), 3) High *TGFB1* and *TGFB3* isoform expression (**Extended data Fig. 7h-j**) and 4) suppressed *TGFBR2* and *TGFBR3* expression (**Extended data Fig. 7k-m**). We note higher variability in the quantification of 3-D co-culture because endothelial cells in non-profiled adjacent sections may influence fibroblast gene expression in the spatially profiled section.

In addition to spatial profiling of 3-D co-cultures we also compared the kinetics of fibrogenic protein production across 2-D and 3-D culture systems (**Extended data Fig. 9a,b**). We quantified daily supernatant Pro-collagen 1A1, POSTN and COMP protein levels, observing highly comparable kinetics across 2-D and 3-D cultures. Co-cultured fibroblasts, compared to monocultured fibroblasts, produced high and sustained Pro-collagen 1A1 and POSTN that was significantly diminished with the use of inhibitors against Notch or TGFβ. Additionally, we observed time-dependent suppression of COMP production in co-culture and monoculture, with similar kinetics in both contexts. Lastly, to further improve the translatability of our *in vitro* mechanistic findings to an *in*

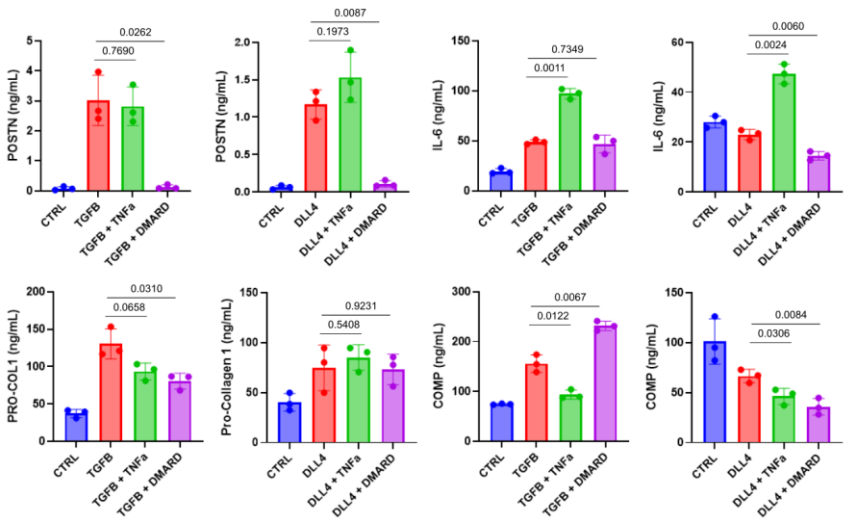


vivo context, we developed a novel RA patient-derived organoid approach (PDO, please see comment **R1C3**) that exhibited the same spatial patterning that we observed in 2-D and 3-D cultures and responded to inhibition of Notch and TGFβ in a comparable manner. <u>Altogether, our profiling of gene and protein expression across 3 model systems supports Notch-mediated tuning of TGFβ signaling as a highly conserved, reproducible mechanism across different experimental models</u>



R3C5 ***"****Additionally, the pre/post treatment synovial biopsy data (figure 7c-d) suggest that DMARD treatment, but not anti-TNF treatment, is associated with an increased percentage of COMPhi fibroblasts in vivo. Do any or all of the DMARDs have direct effects on readouts in the in vitro EC-FB coculture systems (i.e., absent an immune cell component)? Does the presence or absence of TNF in vitro affect Notch or TGFb signaling and/or fibroblast phenotypes?***"*

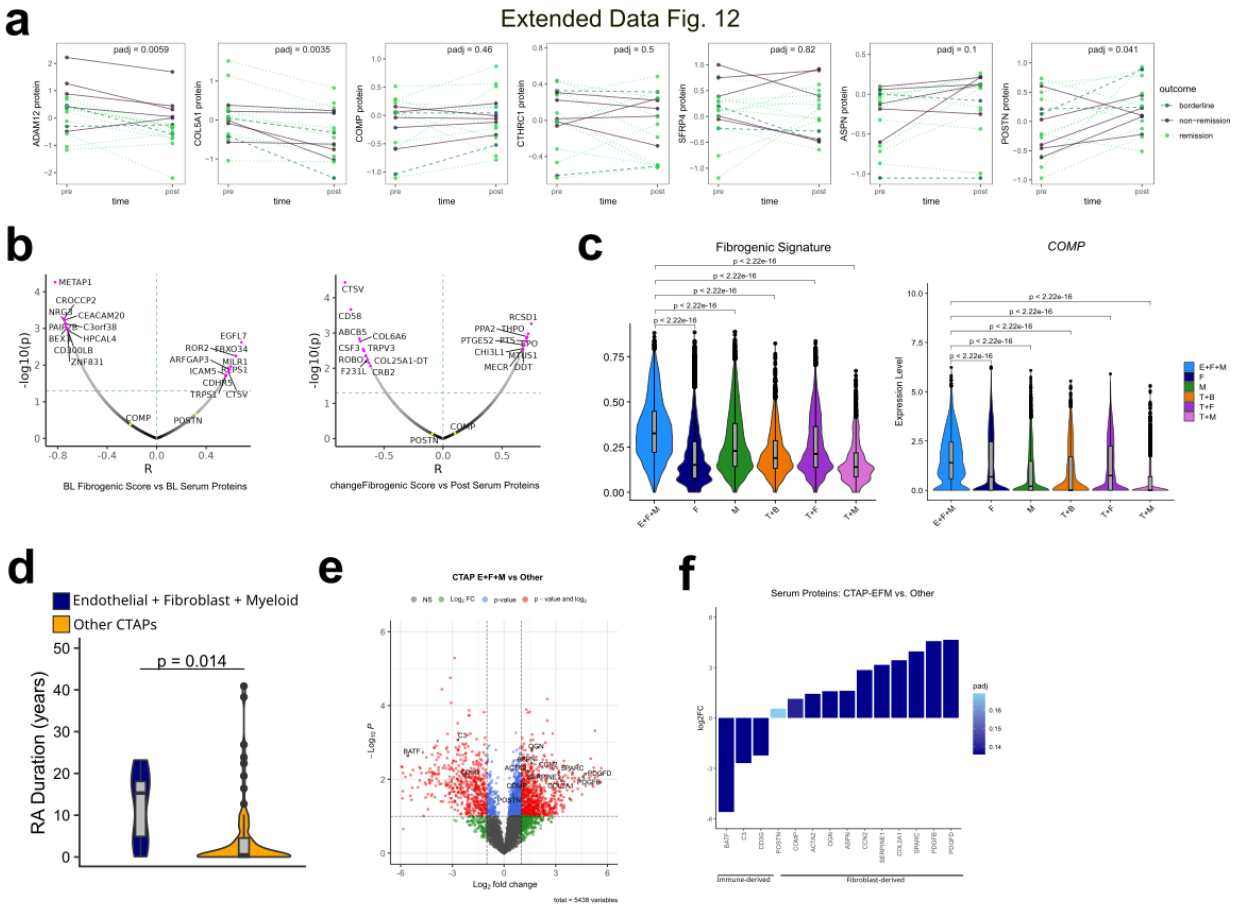
We thank the reviewer for encouraging us to more closely examine the effect of specific treatments on fibrogenic protein production. To directly address the potential role of treatment, we stimulated fibroblasts with either DLL4 (5 ug/mL coated) or TGFβ (1 ng/mL) with or without the addition of DMARDs (methotrexate 10 uM, sulfasalazine 10 uM, hydroxychloroquine 10 uM) or TNFα (1 ng/mL) for 72 hours and assayed the supernatant for POSTN, COMP, Pro-Collagen I and IL-6 (positive control for TNFα stimulation). Our main findings included 1) significant suppression of POSTN production by DMARDs and 2) significantly increased production of COMP with DMARD treatment in the context of TGFβ stimulation, which is consistent with the expansion of COMP+ fibroblasts that was observed specifically in the DMARD group. However, given the highly complex cellular and molecular effects of RA treatments, we did not feel confident about including this data in the manuscript. While we observed in our *in vitro* context that DMARDs may enhance fibroblast COMP production,



we cannot rule out the possibility that the expansion of the COMP+ fibroblast population is downstream of changes in cellular composition or signaling mediated by anti-TNF therapy compared to DMARDs.

R3C6. "*Biomarkers: as noted in comment #2 (above), were serum samples available for blood biomarker analysis in the 396.10 study? If so, are the effects consistent with the data from the AMP study in figure 7i? Olink measures over 5000 protein analytes but it's not clear whether the indicated biomarkers in 7i are the most significantly different between CTAP groups among all the analytes or were cherry-picked to make a particular point (perhaps a volcano plot of all the analytes would be instructive here or in a supplementary figure). Why is periostin not shown? Considering that all the characterization of fibroblast subpopulations in the spatial data and the well-established use of blood periostin as a biomarker in multiple inflammatory and fibrotic conditions (it is also on the Olink panel and present in the linked raw data file), this seems like a curious omission. Given the mechanistic and spatial data elsewhere in this study (discussed in comment #3, above), one could speculate that periostin in POSTNhi fibroblasts is more TGFb3 (+/- TGFb1) dependent whereas COMP in COMPhi fibroblasts is more TGFb2/TGFBR3 dependent. Ultimately, if the findings in this study are to have clinical utility, noninvasive biomarkers will likely be needed to enable and interpret future interventional studies.*"

We thank the reviewer for their suggestion to profile serum from the 396.10 cohort to understand whether spatial or clinical features can be captured in systemically measurable ways. To address this, we performed Olink measurement of serum from the 396.10 cohort and updated the



manuscript with additional analysis (**Extended data Fig. 12a,b**). First, we analyzed individual components of the fibrogenic gene signature, including COMP, in pre- and post-treatment serum. Despite *COMP*+ fibroblast abundance significantly increasing across remission groups, we did not detect any significant changes in COMP at a serum level (padj = 0.46), though we did observe increases in POSTN (padj = 0.041). To determine whether we could identify other serum proteins that can track synovial fibrogenesis at baseline, we correlated patient-level fibrogenic fibroblast score and baseline serum proteins (**Extended data Fig. 12b)**. We identified several correlates such as EGFL7 (r = 0.685, p = 0.0024, padj = 0.76), but no strong pattern of stromal association among the top correlated proteins. Similarly, we evaluated whether changes in fibrogenic signature scores (i.e. increased in fibrogenic score) could be detected in post-treatment serum but found no consistent classification for the top correlated proteins. Comparison of remission versus non-remission serum (data not shown) also did not reveal a classifiable protein profile. To verify whether Olink measurements were capturing inflammatory biomarkers that are expected to be elevated in pre-treatment RA, we correlated baseline disease activity with baseline serum measurements and did indeed find a number of inflammatory biomarkers (CD14, correlation rank = 51, r = 0.57, p = 0.017, padj = 1; C3, correlation rank = 101, r = 0.52, p = 0.032, padj = 1) though notably neither of these markers were among the top 50 correlated proteins. Overall, we believe that we are underpowered with the small size of our cohort to draw conclusions about the ability to systemically detect synovial fibrogenic activity. An additional possibility is that patients in our cohort have too short a disease duration for processes in synovium to be detected systemically. In support of this, we found a strong pattern of stromal-associated protein enrichment (i.e. SPARC, COL3A1, **Extended data fig. 12d-f**) in CTAP-EFM patients who had years-long disease duration. Lastly, we would like to note that we updated the volcano plot and bar plot (**Extended data fig. 12e,f**) to include POSTN and show appropriate adjusted p-values.

R3C7. "*Recommend biostatistical review: in the methods section, it is stated that p-values are not adjusted for multiple testing; given the number of hypotheses tested in several of the experiments, is this appropriate? For example, in the analysis of the Olink biomarker data (see comment #5, above), the "unadjusted" p-value for serum COMP in CTAP-EFM vs. other is about 0.01 according to the color map. Even assuming that the analytes shown were rationally pre-specified and limited, there are 10 of them, thus a simple Bonferroni adjustment would yield a "non-significant" p-value according to the Methods section ("a p-value < 0.05 was considered statistically significant"). In the analyses of spatial data, there are some staggeringly low p-values (e.g., 10e-308 in figure 6n), which may be plausible given that >10e4 cells were imaged; nevertheless, the substantial overlap in distributions calls into question whether these highly "significant" p-values actually represent biologically meaningful differences and whether the statistical methods employed are appropriate for this type of data.*"

We thank the reviewer for their suggestions regarding statistics. For the specific items noted, we have updated the Olink analysis (**Extended data Fig. 7f**, barplot) to show adjusted p-values, but displayed -log10(unadjusted p-values) in the volcano plot (**Extended data Fig. 7e**) for the purpose of visualization. Regarding the quantification of RNAscope data (**Fig. 6**), we acknowledge that while the p-values are low given the high number of cells sampled, all quantified differences that were significant and non-significant were concordant with our visual examination of gene expression patterns within the entire 96-well, which we highlight with representative images. Additionally, we have re-examined our statistical methodology throughout the manuscript to ensure more robustness, which we have highlighted where appropriate.

1. Stein, D. J., Tran, M. & Korsunsky, I. Accurate tiling of spatial single-cell data with Tessera.

bioRxiv (2025) doi:10.1101/2025.01.17.633630.

2. Lewis, M. J. *et al.* Molecular Portraits of Early Rheumatoid Arthritis Identify Clinical and Treatment Response Phenotypes. *Cell Rep.* **28**, 2455-2470.e5 (2019).
3. Acharya, C. *et al.* Cartilage oligomeric matrix protein and its binding partners in the cartilage extracellular matrix: interaction, regulation and role in chondrogenesis. *Matrix Biol.* **37**, 102–111 (2014).
4. Liu, C.-J. *et al.* Transcriptional activation of cartilage oligomeric matrix protein by Sox9, Sox5, and Sox6 transcription factors and CBP/p300 coactivators. *Front. Biosci.* **12**, 3899–3910 (2007).
5. Wei, K. *et al.* Notch signalling drives synovial fibroblast identity and arthritis pathology. *Nature* **582**, 259–264 (2020).
6. Low, S., Barnes, J. L., Zammit, P. S. & Beauchamp, J. R. Delta-Like 4 Activates Notch 3 to Regulate Self-Renewal in Skeletal Muscle Stem Cells. *Stem Cells* **36**, 458–466 (2018).
7. Zhang, C., Weimann, A., Stolzenburg, J.-U., Neuhaus, J. & Berndt-Paetz, M. Notch2/3-DLL4 interaction in urothelial cancer cell lines supports a tumorigenic role of Notch signaling pathways in bladder carcinoma. *PLoS One* **20**, e0317709 (2025).
8. Korsunsky, I. *et al.* Cross-tissue, single-cell stromal atlas identifies shared pathological fibroblast phenotypes in four chronic inflammatory diseases. *Med* **3**, 481-518.e14 (2022).
9. Tefft, J. B. *et al.* Notch1 and Notch3 coordinate for pericyte-induced stabilization of vasculature. *Am J Physiol Cell Physiol* **322**, C185–C196 (2022).
10. Plikus, M. V. *et al.* Fibroblasts: Origins, definitions, and functions in health and disease. *Cell* **184**, 3852–3872 (2021).
11. Sun, T. *et al.* TGFβ2 and TGFβ3 isoforms drive fibrotic disease pathogenesis. *Sci. Transl. Med.* **13**, eabe0407 (2021).
12. Wieteska, Ł. *et al.* Structures of TGF-β with betaglycan and signaling receptors reveal mechanisms of complex assembly and signaling. *Nat Commun* **16**, 1778 (2025).

13. López-Casillas, F., Wrana, J. L. & Massagué, J. Betaglycan presents ligand to the TGF beta signaling receptor. *Cell* **73**, 1435–1444 (1993).
14. Radaev, S. *et al.* Ternary complex of transforming growth factor-beta1 reveals isoform-specific ligand recognition and receptor recruitment in the superfamily. *J Biol Chem* **285**, 14806–14814 (2010).
15. Sun, T. *et al.* Isoform-selective TGF-β3 inhibition for systemic sclerosis. *Med* **5**, 132-147.e7 (2024).

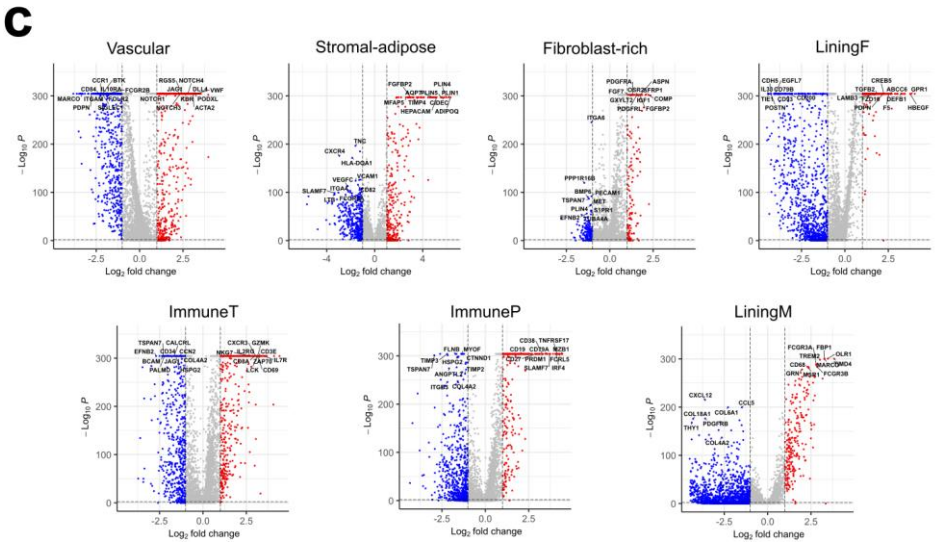
We thank the reviewers for their time and effort in considering our revised manuscript, **NI-A39921A** "**Spatial patterning of fibroblast TGFβ signaling underlies treatment resistance in rheumatoid arthritis**" for publication in Nature Immunology. We are delighted to see that Reviewer 1 (R1) concerns have been "adequately addressed" and that reviewer 2 (R2) found several of the new data "excellent" additions. Further, reviewer 3 (R3) commented that we "have made extensive revisions, included additional information, and added substantial experimental data to strengthen their case for a dysregulated synovial fibroblast population characterized by Notch- and TGFb-dependent signaling gradients in treatment-refractory rheumatoid arthritis." In our revised manuscript, we provide additional analyses and textual amendments to address each remaining point raised by the reviewers. Please see our point-by-point response to the reviewers' comments below.

Reviewer #2

R2C1. ***"****The authors use cell-type agnostic clustering that provides spatial differences in transcript expression. The exact methods used in the clustering and the overall transcriptomic distinctions between clusters are not presented, and this detracts from the primary conclusion. Figure 1b represents the clusters (albeit the resolution and the size of the figure make it difficult to understand the clarity of the cluster differences, although it can be broadly made out). How exactly the 7 niches, namely fibroblast-rich, vascular, stromal-adipose, lining fibroblasts, lining macrophages, T and B cells, and plasma cells, were delineated. A clearer description of this categorization is essential to understanding the validity of this grouping.*

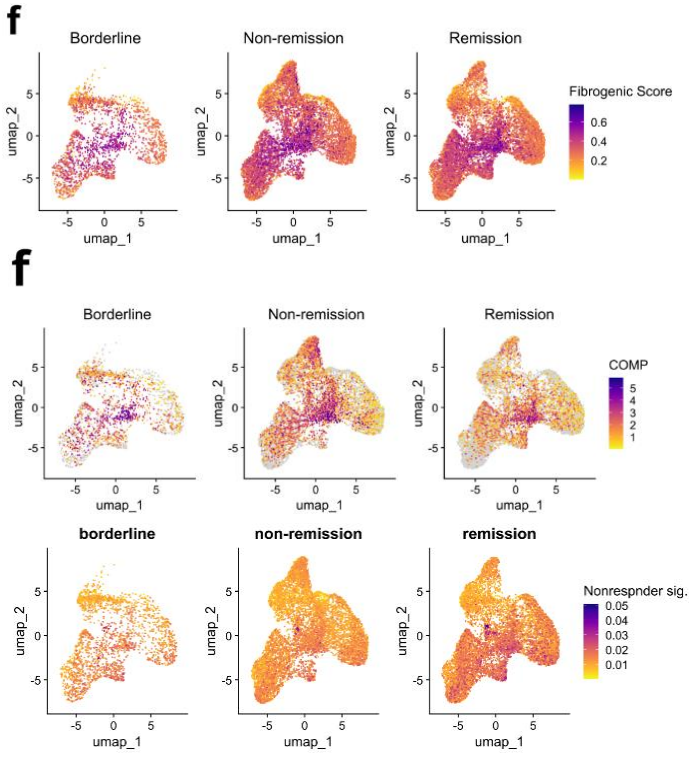
The authors appear not to have done this analysis.

We agree with the reviewer's recommendation to include additional information regarding the methods used for niche identification and their transcriptomic distinctions, which aids interpretation of the data presented. First, we have added important methodological details to the results (lines 127-140). Additionally, we have included volcano plots (**Extended Data Fig. 1c**) that highlight transcriptional identifiers for each of the 7 niches (labeled genes have a padj < 1e-300, Wilcoxon test with Bonferroni correction) and a supplementary file (Table S1) with a comprehensive list of transcriptomic markers for each niche. As can be observed in the additional plots, the transcriptional markers for each niche are consistent with the cell types that compose them: vascular (*VWF, ACTA2)***,** stromal-adipose (*ADIPOQ***,** *PLIN4*), fibroblast-rich (*PDGFRA*, *FGF7*), liningF (*PDPN*, *CREB5*), immuneT (*CD3E*, *IL7R*), immuneP (*CD19*, *MZB1*) and liningM (*CD68*, *FCGR3B*).



R2C2. *"It is more troubling that when the manuscript says, in lines 163-167, that no differences between groups were observed between groups when the baseline and the remission status data were compared. To force the differences, the authors use a bulk transcriptomic data set and filter down a 73-gene signature in non-responders in the bulk study and apply it to the niches. While this may be a reasonable strategy, it is essential to project these signatures across the niches in a UMAP to show that indeed there is a difference between remission and non-remission spatial transcriptomic niches. Without this clear projection UMAP, the conclusions do not make sense."*

We thank the reviewer and agree with their suggestion to project transcript and gene expression signatures onto niche UMAPs to substantiate our conclusions. We have now updated the manuscript to include niche UMAP projections of fibrogenic score (**Extended Data Fig. 3f**) and *COMP* expression (**Extended Data Fig. 4f**) at baseline. The projections show that the 12-gene fibrogenic signature and *COMP* are expressed more broadly across niches in non-remission which complements our quantification of signatures in **Fig. 1h,i**. We also projected the nonresponder signature (shown on the right) which appears limited. However, we have found that the UCell algorithm performs poorly on spatial transcriptomics datasets using large, non-specific gene sets (like the 73-gene signature), because the UCell algorithm penalizes scores per cell if not all genes in the signature are detected, and spatial transcriptomics data has low detection rates compared to scRNA-seq. Many of the genes in the PEAC nonresponder signature are fibroblast-specific, so we were encouraged to see that the non-responder gene signature was enriched in fibroblast niches, consistent with the PEAC authors' previous findings[1]. The limitations outlined above motivated us to finely subtype fibroblasts for the identification of specific non-remission-associated phenotypes.

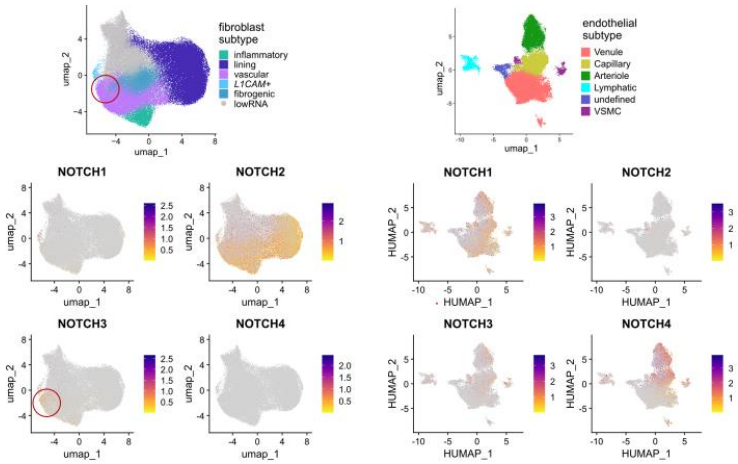


R2C3. *"In a following comment, this reviewer pointed to a question on the vascular niches (see below comments). This is not addressed in the revision.*

There is an interesting marker-based delineation of synovial lining fibroblasts (CD55+) and Notch3+ vascular fibroblasts. In subsequent analysis, where Notch-driven activation of fibroblasts is discussed, it is not clear if the activation is Notch 3, and hence, the fibroblasts are

vascular. The entire Notch discussion in subsequent discussions does not discuss the isoforms.

We agree with the reviewer's recommendation to include additional information regarding Notch isoform expression. Accordingly, in the added volcano plots (**R2C1, Extended Data Fig. 1c**), we have highlighted the relevant Notch isoforms (*NOTCH1, NOTCH3, NOTCH4*) and Notch ligands (*DLL4*, *JAG1*) that are highly enriched in the vascular niche. Further, we examined expression of the receptor isoforms and found that *NOTCH3* expression is specific to mural cell and vascular fibroblast populations and that *NOTCH1* and *NOTCH4* are more highly expressed in endothelial populations (**Figure on right**). We did not include this analysis in the manuscript as it has been well documented in prior literature[2,3].



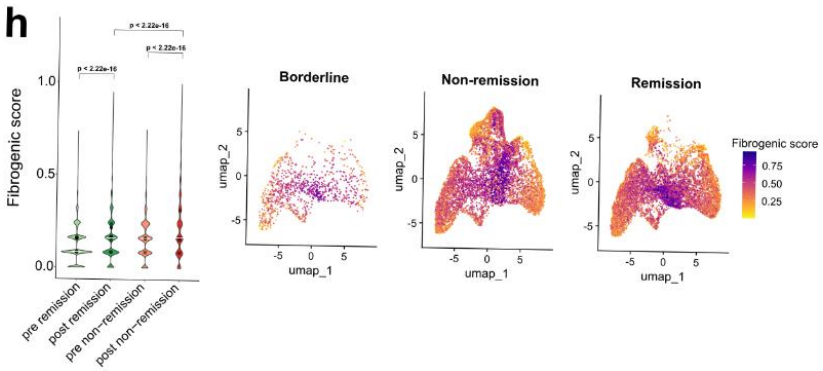
R2C4 *"This reviewer had suggested the analysis of the spatial transcriptomics data from post-treatment biopsies. The authors present only a superficial analysis. Figure 7a, extended data 11b, shows a map post-treatment, which broadly looks like the classification in pre-treatment biopsies. However, as discussed by the authors, "niche composition is affected more by treatment than by remission outcome. This questions the validity of broad biological conclusions drawn on the mechanisms associated with non-remission. This reviewer is less concerned about niche calling differences but deeply concerned about the findings if there are indeed no differences based on remission outcome. Minimally, this warrants clarification and could require more patient data."*

We agree with the reviewer that a deeper analysis of biopsy data is essential for identifying remission-associated phenotypes. First, we would like to acknowledge that sampling contributes to significant heterogeneity among the tissue sections profiled, which can mask differences in niche composition, given the size of our cohort; we have amended the manuscript to describe this limitation (lines 169-172). We are encouraged to observe significant post-treatment contraction of immune niches in our cohort, which is consistent with the effects of immunomodulatory treatment and supports our ability to detect compositional changes in tissue.

In our study we found that the fibroblast phenotype, rather than fibroblast abundance, was associated remission status. The association of cell phenotypes rather than abundance is aligned with prior studies describing pathogenic fibroblast and myeloid subsets in arthritis[4,5].

R2C5 *"One of the important questions that the authors tried to address deals with the increased fibrogenic scores in non-responder patients (with the 10-gene signature). This is not addressed explicitly. Figure 7c shows the fibrogenic scores, but if one looks at the kernel density (color map), there appears to be very little difference. Given the very few non-remission patient data, there should be better statistics. This reviewer had suggested projecting the fibrogenic signatures in the UMAP separately for responders and non-responders to show explicitly that there is indeed a difference. This has not been shown."*

We agree with the reviewers recommendation to show statistics and UMAP representations for fibrogenic score by remission status. We have now included a violin plot representation of fibrogenic score and included statistical comparisons to supplement the kernel density plot (**Extended Data Fig. 11h,** Wilcoxon test with bonferonni correction). Additionally we have included a UMAP representation of fibrogenic score in post-treatment biopsies. We observed that the fibrogenic signature is more broadly expressed across niche tiles in non-remission.



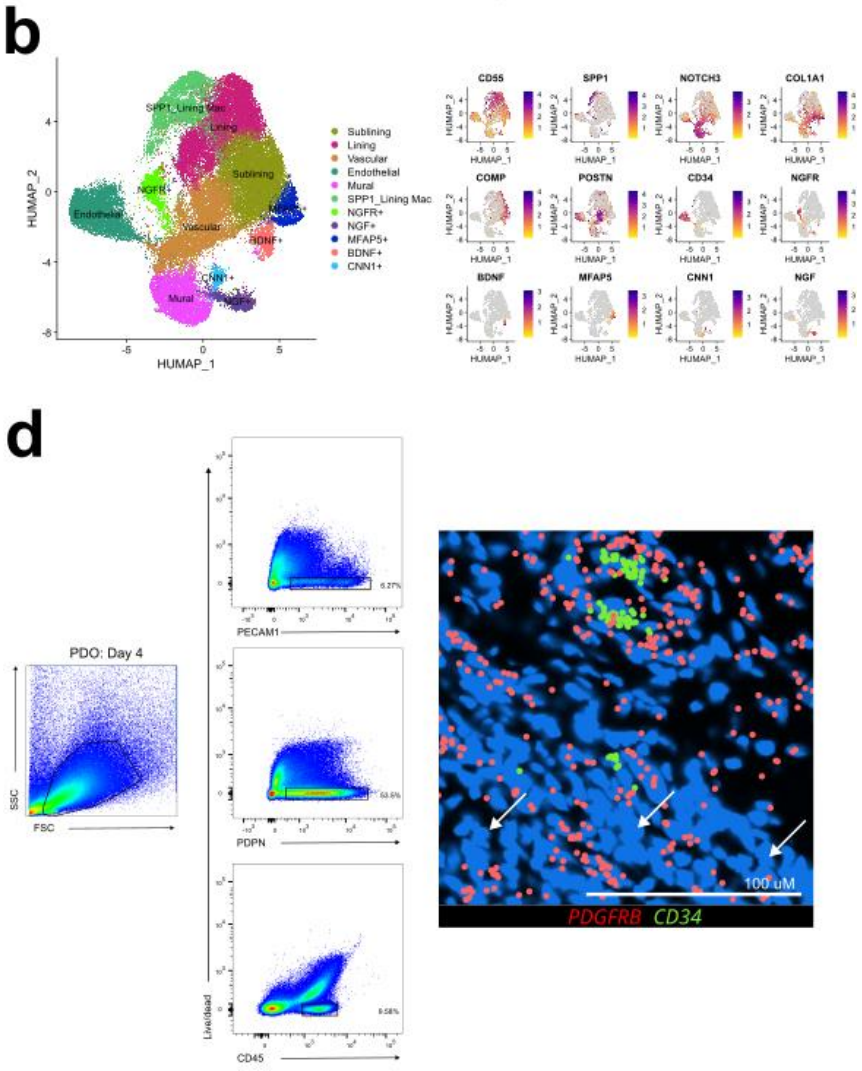
R2C6 "*The analysis and discussion on COMPhi is satisfactory, with a minor caveat that the non-remission patient number is too small to show significant statistics. Is there a way the authors can associate a statistical metric to signify a difference in COMP expression across remission and non-remission patient tissues?"*

In **Fig. 1i** we used the non-parametric Wilcoxon rank-sum test, standard for non-normally distributed data, to quantify baseline differences in normalized *COMP* transcript expression between remission groups (p = 0.075). We are not aware of more appropriate statistical

methods for this comparison and are unfortunately underpowered to reach statistical significance for this specific analysis. We refer the reviewer to additional evidence supporting a role for COMP in non-response to treatment from the PEAC cohort (**Extended Data Figure 4f**). Our lab is currently undertaking spatial analysis of significantly larger RA patient cohorts to validate the association between *COMP* expression and remission status.

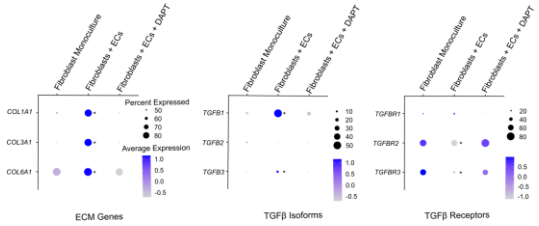
R2C7 *"On author response R2C3 – it is excellent that the authors are using patient-derived synovial organoids. That addresses one question; however, in the R1C3 description, the authors discuss that the organoids only have synovial endothelial cells and fibroblasts. This clearly excludes the immune-associated niche, and the role of a TGFb gradient is unclear given this selective niche."*

We thank the reviewer for their comment and would like to clarify that our patient-derived organoids (PDOs) do contain immune cells that persist in *ex* vivo culture, though our custom fibrogenic gene panel did not contain markers to delineate various immune subtypes apart from *SPP1*+ macrophages. To address this concern, we now include flow cytometric analysis of PDOs cultured at the same time point and conditions as the spatially profiled PDOs (**Extended Data Fig. 13d**). As can be observed in the flow plots, we retain a substantial live CD45+ immune population. Additionally, in our spatial transcriptomic data we observed dense aggregates of nuclei that were negative for fibroblast-specific *PDGFRB* and endothelial-specific *CD34*, suggesting that these may be immune aggregates. Lastly, we would like to note that we did identify an SPP1+ macrophage population, highlighted in **Extended Data Fig. 13b**. We appreciate the opportunity to update our description of PDOs with a discussion of immune populations, which are a highly important feature, per the reviewer's suggestion.



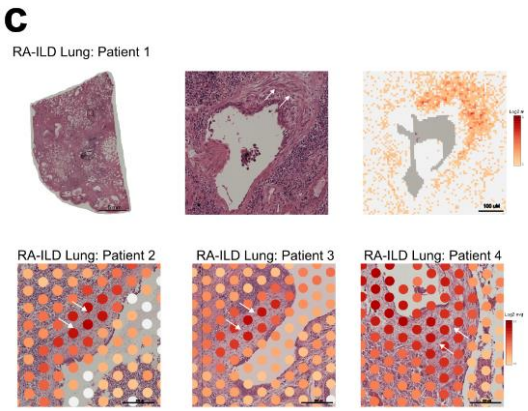
R2C8 *"Response R2C4 is reasonable, but it would help the readers if the authors associated the statistical significance with the dot plots."*

We agree with the reviewer's recommendation to include statistics in association with the dot plots. We have now added asterisks to indicate conditions that had padj values < 1e-100 between fibroblast monoculture and co-culture (**Fig. 4c**, Wilcoxon with Bonferroni correction).



R2C9 *"This reviewer thanks the authors for performing spatial transcriptomics from RA-ILD patient exhibiting usual interstitial pneumonia. This is a good result and hopefully is not unique to the n=1 tissue sample."*

We thank the reviewer for their recommendation to increase our RA-ILD sample size. To address this, we include Visium data for an additional n = 3 RA-ILD explants and have confirmed enrichment of the RA-derived fibrogenic signature in the fibroblastic foci of RA-ILD (**Extended data Fig. 3c**)



R2C10 *"Overall, the manuscript has improved, but some key foundational questions remain. The extrapolation from the authors' spatial transcriptomics data between pre- and post-treatment and those who are non-remission in contrast to remission, warrants more analysis to justify the authors' conclusions. This reviewer refers to the critiques above for the authors to consider."*

We appreciate the reviewer's thoughtful critiques, and believe the additional data and analysis it has inspired strengthens our mansucript.

Reviewer #3

R3C1 *"The additional clinical information in supplementary table 1 and extended data figure 11g are helpful in substantiating that the "responsive" and "refractory" patients had similar aggregate disease activity at baseline. As the authors note in the discussion, while most of these parameters did not reach nominal statistical significance, the sample size was quite small (10 remission vs. 5 non-remission), and it is very interesting to note that the "inflammatory" parameters (e.g., SJC) tended to be higher in responders at baseline while the "pain" parameters (e.g., TJC, VAS pain, PtGA) were higher in non-responders. The changes at 6*

months are also concordant. While I asked for evidence that "non-inflammatory pain" is linked to fibrogenic fibroblast populations and this further evidence is at least consistent with that notion, the manuscript is still missing a mechanistic explanation for how these aberrant fibroblast populations contribute to pain. This could certainly be a topic for further investigation beyond the scope of the present work, but absent mechanistic evidence (e.g., elevated pain mediators, aberrant sensory neurons, etc), it may be more correct to state that there is a link between fibrogenic fibroblast populations and a lack of clinical response in pain parameters to immunomodulatory treatment."

We strongly agree with the reviewer's suggestion to emphasize a link rather than a direct connection between fibrogenic fibroblast populations and pain. We have amended both the introduction (lines 101-105) discussion (lines 726-728) per the reviewer's suggestion.

R3C2 *"Thank you for the additional information on the biopsy collection procedures. I agree that such processes are likely to minimize the potential for sampling bias, but may not completely eliminate it."*

We appreciate the opportunity to update our manuscript with more detailed biopsy methodology and agree that sampling heterogenity likely still exists.

R3C3 ***"****Thank you for the additional experimental data and references on TGFb isoforms, particularly the experiments directed at assessing soluble mediators vs. cell-cell contact. I agree that it is a very complex problem to untangle, particularly with the available tools, and in aggregate, the data suggest that the difference between the proximal and distal fibroblast populations' sensitivity to TGFb is likely more driven by differential expression/availability of TGFb receptors than expression of any particular TGFb isoform."*

We thank the reviewer for suggesting a more detailed examination of TGFβ isoforms and receptors *in vitro*.

R3C4 ***"****The extra effort to reconcile findings in 2D, 3D, and primary "organoid" systems is appreciated. Overall, while specific details vary, the general picture is consistent with the authors' story that Notch signaling proximal to endothelial cells increases periostin, decreases COMP, and decreases the sensitivity of fibroblasts to TGFb via regulation of TGFbRII and/or TGFbRIII."*

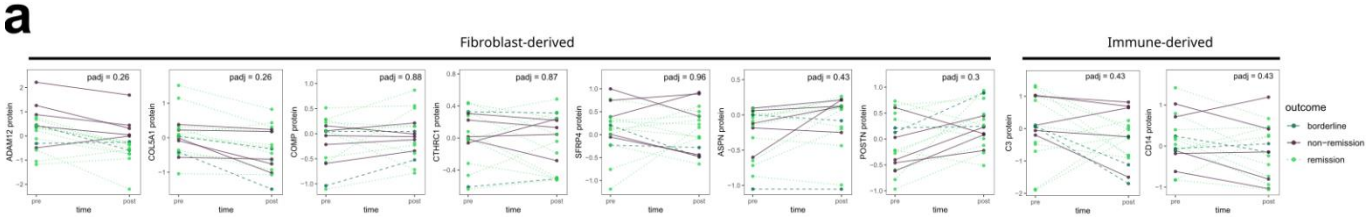
We appreciate the reviewer for encouraging us to explore multiple model systems to dissect mechanisms of Notch and TGFβ signaling which has strengthened our conclusions.

R3C5 *"Thank you for the additional data examining the effects of TNF and DMARDs on key parameters. These findings are interesting but I agree they do not need to be included in the primary manuscript at this time. Eventually, as noted by other reviewers, introducing inflammatory cells into these systems may be necessary to clarify the potential effects of different RA therapies on this milieu."*

We thank the reviewer for their recommendation to consider inflammatory populations. We have added data confirming the presence of CD45+ immune cells in our patient-derived organoids (**R2C7**, **Extended Data Fig. 13d**), though we agree that the development of more sophisticated methods to model inflammation in RA should be addressed in future research. Our group, and others are currently undertaking efforts to more accurately model inflammation in organoid systems.

R3C6 ***"****Thank you for adding biomarker analyses to the 396.10 cohort. As noted, the sample size may be too small to discern meaningful associations with systemic biomarkers; furthermore, as COMP and POSTN in particular are matricellular proteins, the availability of systemically detectable soluble protein may not always align with the expression in the tissue. Finally, the aggregate burden of synovitis across all of the joints in a given patient may further confound the interpretability of systemic biomarkers (e.g., a patient with more active joints may have higher levels than a patient with fewer active joints; adjustment may be necessary for total disease burden at a given timepoint but this study is massively underpowered to answer that). It is nevertheless helpful to be able to visualize COMP and POSTN on the plots and see how they compare to other systemic biomarkers."*

We appreciate the reviewer's comment. To further contextualize the COMP and POSTN results, we have now also included pre and post-treatment quantification of disease activity associated inflammatory markers C3 and CD14 (**Extended Data Fig. 12a**). As can be seen from the plots, C3 shows a trend of decreasing, consistent with the effects of immunomodulatory treatment, but does not reach significance. This emphasizes, as the reviewer suggested, the underpowered nature of our cohort. We and other groups are undertaking serum analysis of larger cohorts to identify response associated biomarkers.



R3C7 *"Thank you for the additional context on statistical methods."*

We appreciate the reviewer for encouraging us to examine our statistical methodology, which we have updated where necessary. More broadly, we found the reviewer's critiques to be very beneficial towards improving our manuscript.

1. Humby, F. *et al.* Synovial cellular and molecular signatures stratify clinical response to csDMARD therapy and predict radiographic progression in early rheumatoid arthritis patients. *Ann. Rheum. Dis.* **78**, 761–772 (2019).

2. Liu, H., Kennard, S. & Lilly, B. NOTCH3 expression is induced in mural cells through an autoregulatory loop that requires endothelial-expressed JAGGED1. *Circ. Res.* **104**, 466–475 (2009).
3. MacKenzie, F. *et al.* Notch4-induced inhibition of endothelial sprouting requires the ankyrin repeats and involves signaling through RBP-Jkappa. *Blood* **104**, 1760–1768 (2004).
4. Alivernini, S. *et al.* Distinct synovial tissue macrophage subsets regulate inflammation and remission in rheumatoid arthritis. *Nat. Med.* **26**, 1295–1306 (2020).
5. Yan, M. *et al.* ETS1 governs pathological tissue-remodeling programs in disease-associated fibroblasts. *Nat. Immunol.* **23**, 1330–1341 (2022).